A source-frozen candidate audit of projection, time-readout, and closure-control kernels in SPARC galaxies

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Abstract

This paper asks whether rotation-curve readout families can be selected from source morphology and projection information before endpoint residuals are inspected. We extend earlier morphology-only kernels to a source-frozen candidate/control framework that separates present morphology, observer projection, morphology-history phase, time-readout controls, and mass/envelope–closure controls. Five source-rich SPARC galaxies are used as a candidate audit: UGC12506, NGC4013, NGC5907, NGC7331, and NGC4088. The audit shows that projection and morphology-history layers can improve simpler morphology-only proxies in selected systems, while remaining neutral or worsening in others. This selectivity is the main result: the added layers do not behave like universal gain knobs. UGC12506 is retained as a stress-test and hypothesis-generation case; its β_{cl} closure expression is a post-diagnostic transfer candidate, not evidence on UGC12506 itself. No population-level gravitational law is validated here. The deliverable is a reproducible internal preflight protocol and a blocked, predeclared validation gate for a future projection-enriched catalogue test.

1 Introduction

Rotation-curve models are often compared through scalar or nearly scalar correction laws: Newtonian baryonic curves, MOND-like interpolations, radial-acceleration relations, or fixed predecessor-readout templates. A morphology-matched forward-readout protocol asks a different question: can source-side morphology select a restricted readout family before the endpoint residuals are inspected?

The one-dimensional version of that protocol is intentionally conservative. A galaxy is assigned a residual-blind readout class, such as scale-tail, exponential disk, compact finite source, ring/resonance, or thick/flared disk. The corresponding kernel is then scored against wrong-family controls and shuffled-label nulls. This paper separates a richer but more demanding layer: observer/projection-enriched readout.

The motivation is simple. The effective readout of an inclined, warped, edge-on, vertically extended, asymmetric, or disturbed galaxy need not be captured by the projected morphology label alone. A source may have the same coarse morphology but present a different line-of-sight projection, warp visibility, vertical overlay, beam/path smearing, morphology-carried trajectory phase, or source-observer geometry. In Tau Core language, this trajectory phase is not limited to past-memory traces: past, present, and future four-dimensional morphologies may be different slices of the same deeper Tau-side configuration. The present paper builds the formula shell and validation gate needed to test that idea without allowing residual leakage into the label. We also isolate the stronger time-readout possibility: projection may modify the effective clock factor used in the observed rotation readout, not merely the spatial morphology kernel.

1.1 Claim boundary

This manuscript is operational. The terms "readout", "projection", and "path" are theory-motivated, but the numerical claim does not require accepting a completed new theory of gravity. The testable object is a source-frozen kernel family selected before endpoint scoring. The current candidate audit is kernel-development evidence only. It motivates, but does not establish, population validation.

1.2 Paper organization

The paper is organized as a source-freeze protocol rather than as a discovery claim. Section 1.3 fixes the distinction between a visible galaxy morphology, a Tau morphology state, observer/path projection, and time-readout projection. Section 2 defines the projection-enriched formula shells and their leakage-prevention requirements. Sections 3–4 report the five-object candidate audit and the diagnostic time/projection ablations. Section 5 isolates UGC12506 as a stress case, while Section 7 keeps its source-derived β_{cl} route explicitly post-diagnostic. Section 6 checks whether the time/projection layer acts selectively across control galaxies. Section 8 then gives the clearest morphology-refinement control ladder, where source review replaces a failed pure beta branch by edge-on vertical/halo kernels before replay. Sections 10–12 state the next population-validation gates.

1.3 Morphology and projection terminology

A central distinction in this paper is that morphology is not identical to projection. The observed four-dimensional morphology of a galaxy is already a projected handle on a deeper source state. It can be used to choose a residual-blind candidate family, but it is not automatically the readout-relevant kernel.

We call the deeper theory-side object a *Tau morphology state*. This is not a visible galaxy morphology class. It is a theory-motivation term for a Tau-side organizing configuration whose four-dimensional readouts may include visible morphology, matter or mass distribution, gravity or metric response, clock/time structure, observer/path appearance, and quantum/coherence structure. The present manuscript does not require accepting a completed model of all these readouts. Operationally, the concept is used only to prevent the visible morphology label from being identified too quickly with the active rotation-curve readout kernel.

In this terminology, the internal Tau morphology state is the common source-side object, while the quantities used below are channel-specific readouts of that object. A present-day visual classification probes one readout; a morphology/gravity kernel probes another; Θ_{morph} probes source trajectory, settling, history, or phase; and Ξ_t or Ξ_{eff} probes the clock/time-slice projection of the observed velocity quotient. These channels can share source evidence because they are readouts of the same deeper state. They cannot, however, be treated as independent endpoint factors unless a residual-blind non-overlap ledger shows that the same source fact is not being counted twice.

The list of channels above is not meant to be exhaustive. In the underlying Tau Core motivation, other projections of the same internal morphology state could also affect the effective rotation readout: for example a mass-distribution readout, a metric/closure readout, a coherence or phase readout, or a source–observer path/environment readout. In this paper those possibilities are kept at the level of protocol discipline. A new projection channel may enter a rotation-curve kernel only if it is converted into a source-frozen observable, assigned to a distinct channel in the ledger, checked against overlap with already active kernels, and tested by ablation before endpoint scoring. Otherwise it remains theory motivation rather than an allowed numerical correction.

A direct consequence is that a Tau Core residual should not be assumed to be locally generated only by the four-dimensional baryonic mass density at the same radius. In the bridge language, the residual can descend from several readout/projection channels of the deeper Tau morphology state: visible baryonic distribution, metric or closure response, observer/path projection, envelope/history/phase structure, or other source-frozen channels. This is not a license to add arbitrary nonlocal forces. Each contributing channel must be source-frozen, assigned in the non-overlap ledger, and ablated before endpoint scoring.

We therefore use four operational objects:

- K_{present}: the present, source-observed four-dimensional morphology handle, for example a disk, warp, bar, ring, flare, compact component, or lopsided feature;
- K_{readout}: the readout-relevant morphology/projection state that is allowed to select the forward formula shell;
- $\mathcal{O}_{\text{obs/path}}$: observer/path projection data, including inclination, line-of-sight stacking, warp visibility, beam/path environment, and source-observer viewing geometry;
- Θ_{morph}: source-frozen Tau-side morphology trajectory or phase information, including history, relaxation, settling, asymmetry, or future-directed phase proxies when independently supported by source data.

The present paper should therefore not be read as assigning final laws to visual classes such as "bar", "ring", or "warp". Those labels are first-pass projected handles. The tested object is a morphology–projection readout family,

$$K_{\text{readout}} = K_{\text{readout}}\left(K_{\text{present}}, \mathcal{O}_{\text{obs/path}}, \Theta_{\text{morph}}, E_{\text{proj/history}}\right),$$

with all arguments frozen from source information before endpoint scoring. This distinction is also what prevents double counting: the same source fact cannot be used once as a morphology kernel and again as an independent projection or clock factor.

After introducing this terminology, we audited the inspected route ledger without recomputing any rotation curve. The audit covers 6 galaxies and 11 routes, including accepted single-object endpoints, null controls, negative controls, and diagnostic time/projection replays. No frozen numerical kernel changes under the terminology update, and no immediate rescoring is required. The audit instead changes the route interpretation: NGC4088 is the resolved case in which the accepted combined route is the additive warp/history endpoint with $\Xi_{\text{eff}} = 1$; UGC12506 remains the main open case where Θ_{morph} and Ξ_t must be separated before any endpoint claim. The subsequent UGC12506 channel gate now performs that separation at the level of formula role: Θ_{morph} is an additive morphology state / trajectory-phase diagnostic, whereas Ξ_t is a multiplicative clock/readout interval-control. This clarifies the architecture, but it does not authorize a combined endpoint because the source context still overlaps. NGC4013's mixed overlay replay remains quarantined as prospective; and NGC7331's broad vertical/outer-warp window remains caveated.

2 Formula shells and source-freeze discipline

2.1 From morphology-only to projection-enriched readout

The morphology-only endpoint shell can be written as

$$\delta v_K^2(R) = A_K\, K_K(R), \tag{1}$$

where K is a residual-blind readout family, $K_K(R)$ is the corresponding radial kernel, and A_K is frozen from source-side rules or a restricted training protocol. Equation (1) is a useful preflight model, but it compresses the observer/projection geometry into a single one-dimensional kernel.

The projection-enriched extension replaces the single kernel by a source-frozen component sum,

$$\delta v^2_{\mathrm{proj}}(R) = \sum_j A_j(\mathrm{source})\, w_j(R, \mathcal{O}_{\mathrm{obs/path}}, \Theta_{\mathrm{morph}})\, K_j(R; K_{\mathrm{present}}, \Theta_{\mathrm{morph}}). \tag{2}$$

Here $K_j(R; K_{\mathrm{present}}, \Theta_{\mathrm{morph}})$ are morphology components selected from the present projected source handle and, where source-supported, the Tau-side trajectory/phase profile; $A_j(\mathrm{source})$ are source-side amplitudes; and $w_j(R, \mathcal{O}_{\mathrm{obs/path}}, \Theta_{\mathrm{morph}})$ are observer/projection visibility weights that may also depend on the trajectory/phase profile. The symbol $\mathcal{O}_{\mathrm{obs/path}}$ denotes source-observer geometry: inclination, position angle, line-of-sight projection, warp visibility, finite resolution, waveband, and velocity-field coverage. All fields must be frozen from source data, not from endpoint residuals.

2.2 Operational derivation of a concrete readout kernel

The concrete galaxy kernels used below are not introduced as free curve templates. Each kernel is derived by the same four-step operational recipe. First, a residual-blind source review selects the active readout channel: present morphology, observer/path projection, morphology-history phase, time/clock readout, mass/envelope closure, or an explicitly mixed channel. Second, that channel is converted into a dimensionless radial support profile $K_j(R)$ with a frozen onset, width, sign, and saturation rule. The rule is chosen from the source statement: for example a published warp onset becomes a smooth radial activation beginning at that radius; an edge-on vertical/halo component becomes a disk–halo interface gate; and a compact or envelope component becomes a bounded support profile rather than a global multiplier. Third, observer-facing geometry supplies only the visibility or projection weight w_j, such as edge-on overlay, warp visibility, or projection saturation. Fourth, the amplitude A_j is either source-normalized or frozen by a predeclared carrier rule before endpoint scoring. If any of these ingredients is chosen after reading the rotation residual, the route is demoted to a diagnostic or control.

Source statement	Kernel ingredient	Typical formula role	Status guardrail
Warp or outer vertical onset	Frozen radial activation $K_{\mathrm{warp}}(R)$	$\delta v^2 = AK_{\mathrm{warp}}$ or mixed projection weight	onset and width must come from source morphology, not residual shape
Edge-on vertical/halo or dust/H I layer	Disk–halo interface gate $K_{\mathrm{EVH/DVH}}(R)$	spatially gated carrier or closure transfer	pure global transfer is kept as a negative control if it overshoots
High-spin, diffuse envelope, or NFW/HSE preference	Mass/envelope closure factor or carrier support	$v^2 = \beta_{\mathrm{cl}} v^2_{\mathrm{carrier}}$ or source-native closure replay	post-diagnostic normalizations require independent transfer before endpoint use
Asymmetry, disturbance, or settling phase	Θ_{morph} trajectory/phase support	additive morphology-state phase correction	may not also count as a clock term unless a non-overlap ledger leaves a remainder
Independent clock/readout evidence	$\Xi_t(R) = 1 + \epsilon_t K_t(R)$	multiplicative time-readout interval	requires source-token priority and an accepted ϵ_t origin

Table 1: Operational derivation recipe for concrete readout kernels. The table explains how source statements are translated into radial kernels, weights, or carrier factors. It is not a list of validated laws: every row still requires source freeze, non-overlap checks, and endpoint-blind scoring before it can be used as endpoint evidence.

2.3 Morphology-dependent observer/path projection

The observer/projection factor in Eq. (2) is not a universal inclination correction. In the Tau-readout interpretation used here, the observed kernel is allowed to depend on four source-side ingredients,

$$K_{\mathrm{readout}} = K_{\mathrm{readout}}\left(K_{\mathrm{present}}, \mathcal{O}_{\mathrm{obs/path}}, \Theta_{\mathrm{morph}}, E_{\mathrm{proj/history}}\right). \tag{3}$$

Here K_{present} denotes the present-day projected source morphology; $\mathcal{O}_{\mathrm{obs/path}}$ denotes the observer/path projection data such as inclination, edge-on overlay, warp visibility, beam/path smearing, and the source-observer viewing geometry; Θ_{morph} denotes the Tau-side morphology trajectory or phase profile, including source-frozen proxies for past traces, present structure, and future-directed relaxation or evolutionary tendency; and $E_{\mathrm{proj/history}}$ denotes the environmental or path-history information that can affect how the source is projected to the observer. Equation (3) is a selection discipline, not an endpoint fitting equation. Each argument must be promoted from residual-blind source evidence before the rotation-curve endpoint is scored.

This distinction matters for edge-on and disturbed systems. Two galaxies can share the same coarse present-day class, for example “thin disk” or “warp”, while having different observer-facing readouts because the warp is seen at a different angle, the vertical component is overlaid differently, or the relevant morphology is a trajectory-bearing projection rather than only the current optical shape. In Tau Core language, future-directed morphology or path-environment terms are not treated as backward causal influences. They are treated as trajectory/phase components of a deeper Tau-side configuration, whose four-dimensional past, present, and future appearances may be different readout slices. Thus a source-frozen relaxation direction, environmental phase indicator, or path-trajectory proxy can be part of the current readout state without being interpreted as a future four-dimensional event acting backward on the present rotation curve. Conversely, image-plane neighbours or apparent interlopers are not automatically path terms. They enter the kernel only if source evidence establishes that they affect the galaxy-to-observer null-geodesic bundle, the foreground metric/lensing environment, or a genuine source-trajectory component. Otherwise they remain masks, caveats, or rejected projection inputs.

A still fuller path-aware kernel would include

$$\delta v^2_{\rm full}(R) = \delta v^2_{\rm proj}(R) + \delta v^2_{\rm path}(R; \mathcal{P}), \tag{4}$$

where $\mathcal{P}$ is the source-observer null-geodesic bundle and the metric or matter environment that can causally affect that bundle. This wording is deliberate: the relevant environment is not every object in a naive spatial cone, but the geometry and matter distribution that influence the light bundle by which the galaxy is observed. The path term in Eq. (4) is not modeled or scored in the present paper. The current audit is therefore projection-enriched, not fully path-aware.

2.4 Projection-channel taxonomy used in this paper

The word "projection" is used in several related but distinct senses. To avoid conflating them, Table 2 summarizes the channels used in this paper, their operational role, and their present claim status. The table is a map of the readout architecture, not a list of validated endpoint effects. A channel becomes endpoint-active only when its source tokens are frozen before scoring and do not double-count a more direct channel.

Channel	Meaning	Examples in this paper	Status	Guardrail
Present morphology readout	Projected source structure selecting the base morphology kernel	NGC4013, NGC7331, NGC5907, NGC4088, UGC12506	main operational layer	source-selected before scoring; not inferred from residuals
Observer/path projection	How the source is read from the observer's line of sight or path geometry	NGC5907, NGC4013, NGC7331, UGC12506	partly endpoint-active in mixed/projection routes	image-plane coincidence is insufficient; source/path support required
Morphology-history / trajectory phase	Source-side disturbance, relaxation, interaction, or phase state carried by morphology	NGC4088, UGC12506, NGC7331	caveated endpoint or control depending on source freeze	future-directed terms are trajectory/phase, not backward causation
Time / clock readout projection	Effective time-slice or clock-readout mismatch in the velocity readout	NGC4088 and UGC12506 controls; NGC4013, NGC5907, NGC7331 guardrails	control/diagnostic in this paper	requires independent clock evidence and nonzero T/A remainder
Path/environment projection	Null-geodesic bundle and metric/matter environment affecting the observed path	UGC12506 path audit; NGC4088 path inactive	protocol/future full-kernel layer	must affect the source-observer bundle, not merely appear nearby
Mass/envelope / closure readout	Deeper source-envelope or closure channel not reducible to local 4D baryonic density alone	UGC12506, NGC0891, NGC4217, IC4202	mostly control/prospective	requires source-native carrier/amplitude freeze; not a curve-rescue term

Table 2: Projection/readout channels used in Paper 2. The table clarifies where each channel appears and which channels are endpoint-active, diagnostic, or future/protocol-only. The released artifact is `projection_channel_taxonomy_*`.

2.5 Time-readout projection channel

The trajectory/phase layer above should not be read only as a morphology label with memory. The stronger Tau Core interpretation is that the observed four-dimensional clock readout can itself be projection-dependent. The rotation speed inferred from a source is not only a gravity-response readout; it is also a quotient of a spatial readout and a time readout. If the projection-selected clock slice differs from the Newtonian closure-test clock slice, the observed velocity can acquire a multiplicative time-readout factor. We write the first shell as

$$v^2_{\rm obs}(R) = \Xi^2_t(R; \mathcal{O}_{\rm obs/path}, \Theta_{\rm morph}, E_{\rm proj/history}) \left[v^2_{\rm Newt}(R) + \delta v^2_{\rm grav/morph}(R)\right], \tag{5}$$

where $\Xi_t = 1$ is the Newtonian clock-readout limit and $\delta v^2_{\rm grav/morph}$ denotes the morphology/gravity residual generated by the readout family. For small projection mismatch, $\Xi_t(R) = 1 + \epsilon_t(R)$, this gives

$$\delta v^2_t(R) \simeq 2\epsilon_t(R) \left[v^2_{\rm Newt}(R) + \delta v^2_{\rm grav/morph}(R)\right]. \tag{6}$$

Equations (5)–(6) are formula shells, not fitted laws. The factor Ξ_t must be frozen from source-side observer/path, morphology-trajectory, or clock-readout evidence before scoring. It cannot be inferred from the rotation residual. In the present paper the full-time morphology replay is therefore only a diagnostic proxy for this channel; a source-complete time-projection endpoint would require an explicit residual-blind $\Xi_t(R)$ manifest.

The channel acts on several projections at once. It does not replace the morphology or gravity kernel; it changes how those kernels are read through an effective clock slice. Table 3 summarizes the operational decomposition.

For clarity, the single symbol Ξ_t in Eq. (5) should be read as an effective compressed factor, not as the final fundamental decomposition. A fuller Tau Core time-projection shell separates at least a source-morphology clock channel, an observer/path clock channel, and a possible path-environment channel:

$$\Xi_{\rm eff}(R) = \Xi_{\rm morph}(R; \Theta^\tau_{\rm src})\Xi_{\rm obs}(R; \mathcal{O}_{\rm obs/path})\Xi_{\rm path}(R; E_{\rm proj/history}). \tag{7}$$

The same time-projection data may also deform the morphology kernel itself,

$$K_{\rm readout}(R) = K_0(R; K_{\rm present}) + \delta K_{\rm morph/time}(R; \Theta^\tau_{\rm src}) + \delta K_{\rm obs/time}(R; \mathcal{O}_{\rm obs/path}). \tag{8}$$

Thus the current diagnostic Ξ_t runs are not tests of the full fundamental time-projection branch. They instantiate a narrow proxy or control slice. In particular, the UGC12506 caveated control partially covers the high-spin/envelope source-morphology clock channel and the edge-on observer/PV clock-slice channel, sets the path term to zero, and does not yet include the kernel-deformation channel of Eq. (8).

Projection subchannel	Formula effect	Residual-blind freeze input
Observer/path projection	sets $\mathcal{O}_{\rm obs/path}$ in Ξ_t	inclination, edge-on overlay, warp visibility, beam/path audit
Morphology trajectory/phase	sets $\Theta_{\rm morph}$ in Ξ_t	settling state, warp/asymmetry stage, interaction or relaxation proxy
Gravity/readout projection	$\Xi^2_t[v^2_{\rm Newt} + \delta v^2_{\rm grav/morph}]$	separately frozen morphology/gravity residual shell
Clock-rate/time-slice projection	$\Xi_t = 1 + \epsilon_t$, with $\delta v^2_t \simeq 2\epsilon_t(\cdots)$	source-frozen clock/readout mismatch proxy
Path/environment projection	possible $E_{\rm proj/history}$ dependence	null-geodesic bundle environment; reject image-plane coincidence alone

Table 3: Projection subchannels affected by the time-readout factor. The table is a source-freeze checklist, not a claim that any listed observable has already been promoted for the inspected galaxies.

For later endpoint work the intended ablation ladder is

$$K^{(0)}(R) = K(R; K_{\text{present}}), \tag{9}$$
$$K^{(1)}(R) = K(R; K_{\text{present}}, \mathcal{O}_{\text{obs/path}}), \tag{10}$$
$$K^{(2)}(R) = K(R; K_{\text{present}}, \mathcal{O}_{\text{obs/path}}, \Theta_{\text{morph}}), \tag{11}$$
$$K^{(3)}(R) = K(R; K_{\text{present}}, \mathcal{O}_{\text{obs/path}}, \Theta_{\text{morph}}, E_{\text{proj/history}}). \tag{12}$$

The curves already plotted in this paper should be read as $K^{(0)}$, $K^{(1)}$, or partial $K^{(2)}$ approximations, depending on the source artifact. They are not full $K^{(3)}$ path-aware kernels.

2.6 Source-token priority between morphology and time projection

The time-projection channel creates an additional leakage risk: the same source datum can sometimes be described both as a morphology/projection indicator and as a clock/readout indicator. The protocol therefore assigns source tokens before endpoint scoring. The rule is not chosen by which route gives the better rotation curve.

The priority rule is:

1. A token that directly describes shape, geometry, warp, asymmetry, bar/ring structure, vertical overlay, or observer/path appearance is first assigned to the morphology or observer-projection kernel.
2. A token may enter the time-projection channel only if it carries independent clock/readout content, such as component settling, relaxation state, multi-waveband phase offset, lag/clock mismatch, or path-clock evidence not already consumed by the morphology kernel.
3. One source token may contribute to only one endpoint channel. If both readings remain plausible, the time contribution is measured only by the quotient remainder T/A, where A is the active morphology/projection source subspace and T is the candidate time-projection source ledger.
4. If no independent separator exists, the lower-level direct morphology/projection reading wins, while the time reading may remain a diagnostic or control route.
5. Rotation residuals, RMSE, wrong-family rank, or endpoint performance may not break the tie.

This is the protocol used in the NGC4088 non-overlap certificate below. Warp onset, warp strength, orientation mismatch, and morphology-history phase are first assigned to the accepted warp/history morphology route. The current Ξ_t ledger has no quotient-surviving remainder on top of that route, so the time-projection curve remains a control even though it improves the RMSE. The released artifact is `projection_channel_priority_protocol_*`.

2.7 Source-freeze requirements

Projection enrichment is scientifically useful only if the additional labels are frozen independently of endpoint residuals. A source-frozen projection manifest should include, where available:

- inclination, position angle, and line-of-sight geometry;
- H I extent, warp onset, tilted-ring structure, and deprojection constraints;
- vertical scale height, flare, extra-planar gas, and edge-on overlay evidence;

- bar, ring, lopsidedness, asymmetry, or interaction/history markers;
- relaxation direction, settling phase, accretion/stripping stage, or other source-side proxies for future-directed morphology phase;
- time-readout or clock-projection proxies, where available, such as path-dependent redshift/deprojection consistency, velocity-field time-slice coherence, relaxation/settling clock indicators, and environment-supported clock/readout mismatch evidence;
- waveband and resolution metadata relevant to the observed morphology;
- a label-freeze artifact recorded before endpoint scoring.

Forbidden endpoint fields include the final rotation-curve residuals, the identity of the best-fitting wrong family, and any post-hoc kernel choice made after inspecting the endpoint score. A future-directed morphology proxy is allowed only when it is inferred from residual-blind source information, such as geometry, H I asymmetry, tidal context, stellar-population gradients, or resolved velocity-field structure; it cannot be inferred from the residual shape that the kernel is supposed to predict.

3 Candidate sample and audit

3.1 Candidate ledger

Table 4 lists the five source-rich systems used in the current candidate audit. They were chosen because independent literature provides projection-relevant morphology information, not because they prove a population result.

Galaxy	Projection handle	Readout motivation	Representative source support
UGC12506	edge-on high-spin H I envelope plus projection-history caveat	edge-on PV/envelope structure, extended diffuse H I, high spin, and asymmetry motivate a caveated projection-history lane	H I envelope and kinematic context [7]
NGC4013	warp plus vertical overlay	edge/vertical structure can change the effective radial readout relative to a simple disk proxy	H I lag/warp context [21]; vertical structure support [3]
NGC5907	edge-on projection plus warp/truncation	line-of-sight stacking and outer warp make observer geometry readout-relevant	warp evidence [17, 18]; edge-on ISM context [20]
NGC7331	vertical/outer warp overlay	outer H I and vertical structure motivate a mixed outer-readout lane	H I/outer context [1]; scale-height analysis [16]
NGC4088	asymmetric warp/trajectory projection	strong disturbance and asymmetry motivate a trajectory/projection lane	Ursa Major H I synthesis and asymmetry context [19]

Table 4: Projection-enriched candidate ledger. The entries are source-side motivations for projection labels, not endpoint-fitted explanations.

3.2 Projection-enriched candidate audit

The candidate audit compares a simpler morphology-only proxy with the best available source-frozen observer/projection or morphology-trajectory enriched Tau curve for each inspected object. The purpose is to test whether projection enrichment is worth a future catalogue campaign. It is not population validation and it is not a final path-aware kernel. The inspected systems were not selected as a statistically independent validation sample.

Galaxy	Simple	Obs./hist.	Best base	Gain	Caveat
UGC12506	102.48	69.18	38.12	33.30	caveated control replay; still far from MOND/TPG diagnostics
NGC4013	10.88	10.61	12.27	0.27	retrospective mixed-reference case
NGC5907	17.03	15.50	16.79	1.53	observer projection beats mixed curve; tight control margin
NGC7331	23.47	22.13	25.49	1.34	source-sharpened outer-warp replay caveat
NGC4088	38.46	11.62	25.40	26.84	source-review and normalization caveats

Table 5: Candidate audit RMSE values in $\mathrm{km\,s^{-1}}$. "Simple" is the simple Tau morphology proxy, "Obs./hist." is the best available observer/projection or morphology-trajectory enriched Tau curve in the current artifacts, and "Best base" is the best listed non-Tau comparator available in the same point artifacts. UGC12506 remains a caveated control replay and NGC4013 is a retrospective mixed-reference case; the table is kernel-development evidence, not population validation.

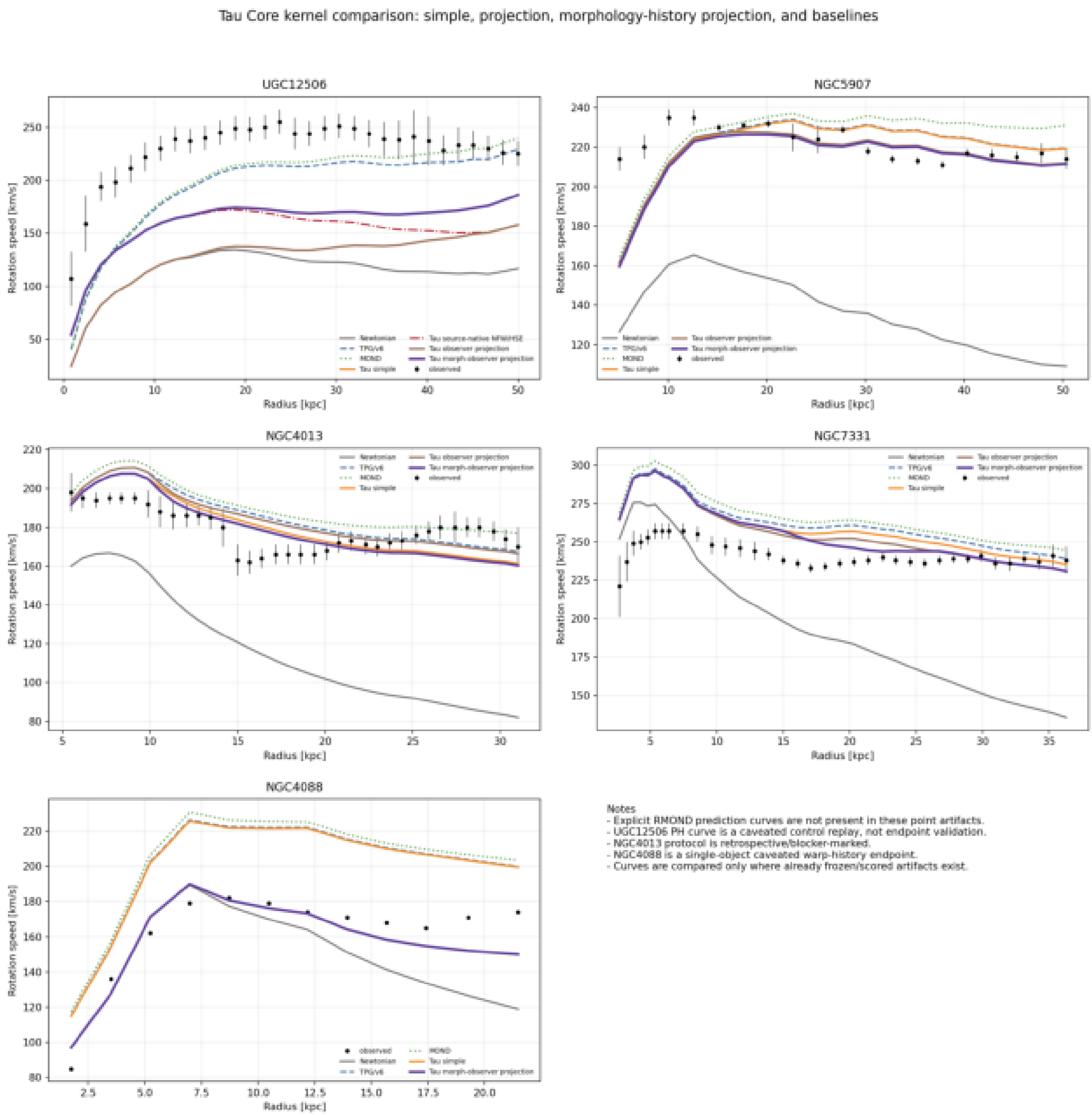


Figure 1: Projection-enriched rotation-curve comparison for the five inspected candidates. The plotted curves show the measured rotation curve, Newtonian baryonic comparator, TPG/v6 predecessor comparator, MOND comparator where available in the point artifacts, the simple Tau morphology proxy, and the observer/projection or morphology-trajectory enriched Tau variants. An explicit RMOND prediction curve is not present in the current point artifacts and is therefore not drawn. The figure is intended as kernel-development evidence: it illustrates how the enriched kernel differs from the simpler morphology-only proxy and from listed comparators. It is not a population endpoint figure.

The audit is consistent with the expectation that observer/projection and morphology-trajectory information is not a negligible correction to morphology-specific readout kernels. UGC12506 is the clearest stress case: the simple envelope and observer-projection shells remain weak, the source-native NFW/HSE shell improves the result, and the incremental projection-history shell improves further, but the object still does not reach the strongest MOND/TPG diagnostic curves. This is evidence that projection-history matters, not an endpoint success. In NGC5907 this is natural because the system is edge-on and projection-dominated. In NGC4013, warp and vertical overlay make a mixed readout plausible. In NGC7331, the improvement is smaller and caveated by outer-warp source-freeze choices. In NGC4088, the improvement is large, but the case is also the most source-review-sensitive because asymmetric history and normalization choices matter.

4 Time/projection diagnostic layers

4.1 Full-time morphology phase diagnostic

The preceding comparison uses the strongest available projection-enriched or trajectory-enriched curve already present in the object-specific artifacts. As a separate diagnostic, we also ran a small "full-time morphology" phase layer on the trial galaxies for which point-level projection artifacts are available. In Tau Core language this is not a backward-causality term. It is a trajectory/phase component of the deeper morphology configuration, whose four-dimensional present, past, and future appearances may be different readout slices. Operationally, the trial layer has the form

$$v_{\mathrm{FT}}^2(R) = v_{\mathrm{base}}^2(R) + A_{\mathrm{FT}}(\mathrm{source})\, K_{\mathrm{FT}}(R), \tag{13}$$

where v_{base} is the already frozen morphology/projection curve, K_{FT} is a late-settling trajectory/phase kernel constructed from the existing source-side projection kernel and radial support, and A_{FT} is fixed from the source-status load and the existing carrier scale. The observed rotation curve is used only for scoring. The trial is not an endpoint validation and is not a final path-aware kernel. It is also not a complete time-projection endpoint in the sense of Eq. (5); it is a first diagnostic proxy for the possibility that projection-selected morphology phase and clock/readout phase modify the inferred rotation together.

Galaxy	Base	Full-time	Difference	Interpretation
NGC4013	11.45	11.51	+0.06	slight degradation; existing warp/vertical-overlay curve is already close to the accepted shape, so the added phase layer over-counts source structure already carried by the mixed kernel
NGC4088	11.62	9.39	−2.23	clear improvement in the strongest warp/history/asymmetry case
NGC4183	6.25	6.25	−0.00	null-like response, as expected for a weak-projection control case
NGC5907	15.50	15.50	−0.00	nearly saturated projection kernel; the phase layer is effectively neutral
NGC7331	22.26	22.28	+0.02	slight degradation; broad mixed/outer-warp kernel may already include the available phase information, so the phase proxy adds a diffuse correction rather than a sharper independent source component

Table 6: Diagnostic full-time morphology phase replay on trial galaxies. RMSE values are in $\mathrm{km\,s^{-1}}$. Negative differences mean that the full-time phase layer improved the existing source-frozen morphology/projection curve. The construction is source-frozen and diagnostic only; no endpoint validation claim is made.

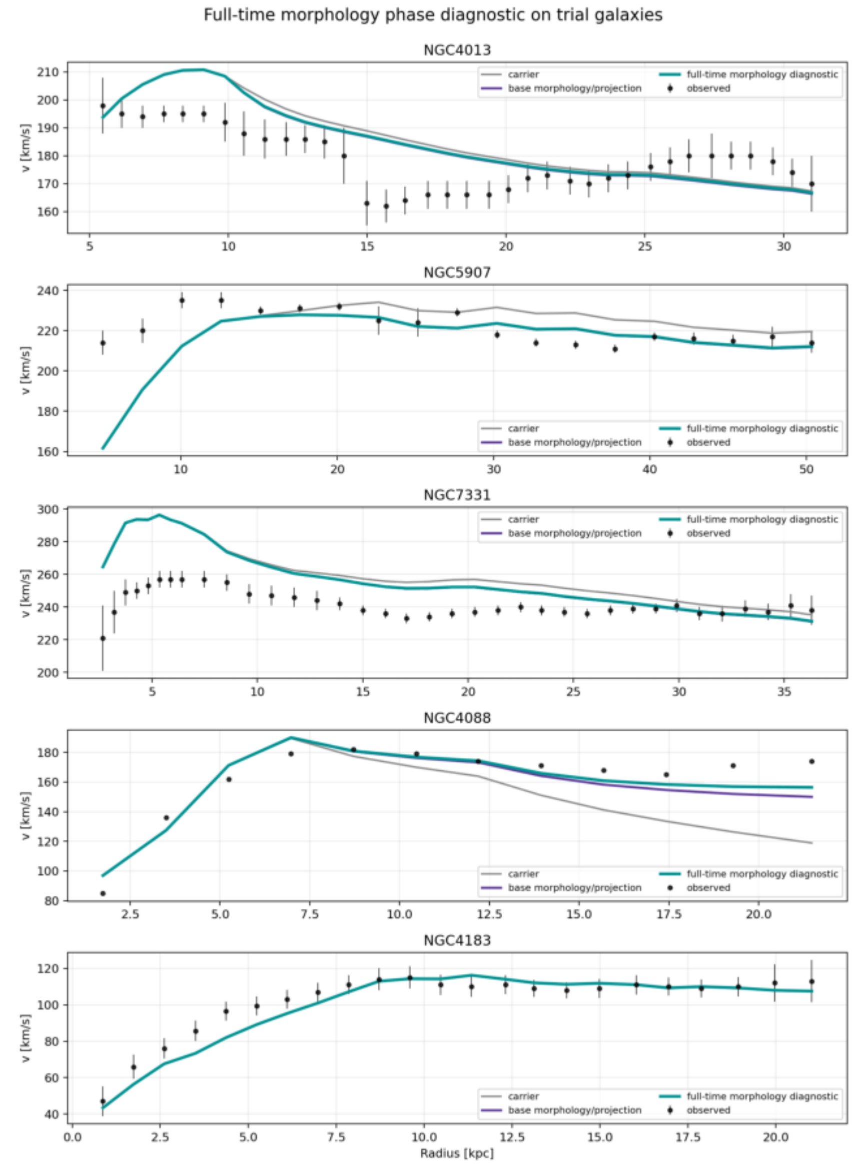


Figure 2: Full-time morphology phase diagnostic on the trial galaxies. The purple curve is the already frozen morphology/projection baseline and the teal curve adds the diagnostic trajectory/phase layer of Eq. (13). The layer gives a meaningful improvement only for NGC4088, the strongest warp/history/asymmetry object in the current trial set. It is nearly neutral for NGC5907 and NGC4183 and slightly worsens NGC4013 and NGC7331. This pattern is useful because the phase layer does not behave as a universal gain knob; its effect is morphology/history dependent.

The diagnostic result is therefore claim-safe but informative. The full-time morphology phase layer is not a universal additive correction that improves every galaxy by construction. It helps most where the source status indicates strong time/trajectory, warp, or asymmetry content, as in NGC4088. In more stable systems, or in systems where the projection/mixed kernel is already close to saturated, the additional layer is negligible or slightly harmful. This is consistent with the Tau Core expectation that trajectory/phase information should be readout-relevant only for the morphology classes that carry such information.

4.2 Diagnostic Ξ_t replay

To separate the additive morphology-phase proxy from the explicit clock/readout shell of Eq. (5), we ran one additional diagnostic. The replay applies a small residual-blind factor $\Xi_t(R) = 1 + \epsilon_0 K_t(R)$ to the existing base and full-time curves. The coefficient ϵ_0 is capped and derived from the same source-status load used by the trajectory/phase replay. It is not fitted from the rotation residual. This test asks only whether the time-readout direction is plausible enough to motivate a future accepted $\Xi_t(R)$ manifest.

Galaxy	Base	Additive	Ξ_t+base	Ξ_t+add.	Interpretation
NGC4013	11.45	11.51	11.95	12.04	clock factor worsens an already good mixed/overlay curve; current clock proxy is not independent of the active warp/vertical-overlay source context
NGC4088	11.62	9.39	10.50	8.38	clear improvement in strong warp/history/asymmetry case
NGC4183	6.25	6.25	6.24	6.24	nearly null response, consistent with weak-projection control status
NGC5907	15.50	15.50	15.52	15.53	slight degradation; projection kernel appears saturated
NGC7331	22.26	22.28	22.41	22.44	slight degradation; broad mixed outer-warp curve already carries the signal, so the clock factor over-rescales a saturated broad-window proxy
UGC12506	69.18	64.12	67.90	62.97	improves the problematic high-spin edge-on stress case, but remains diagnostic

Table 7: Diagnostic Ξ_t replay on trial galaxies. Values are RMSE in $\mathrm{km\,s^{-1}}$. "Additive" is the full-time morphology phase proxy of Eq. (13); the last two columns multiply the base or additive curve by the small source-frozen clock/readout factor. The replay is not an accepted endpoint because no residual-blind $\Xi_t(R)$ manifest has yet been promoted.

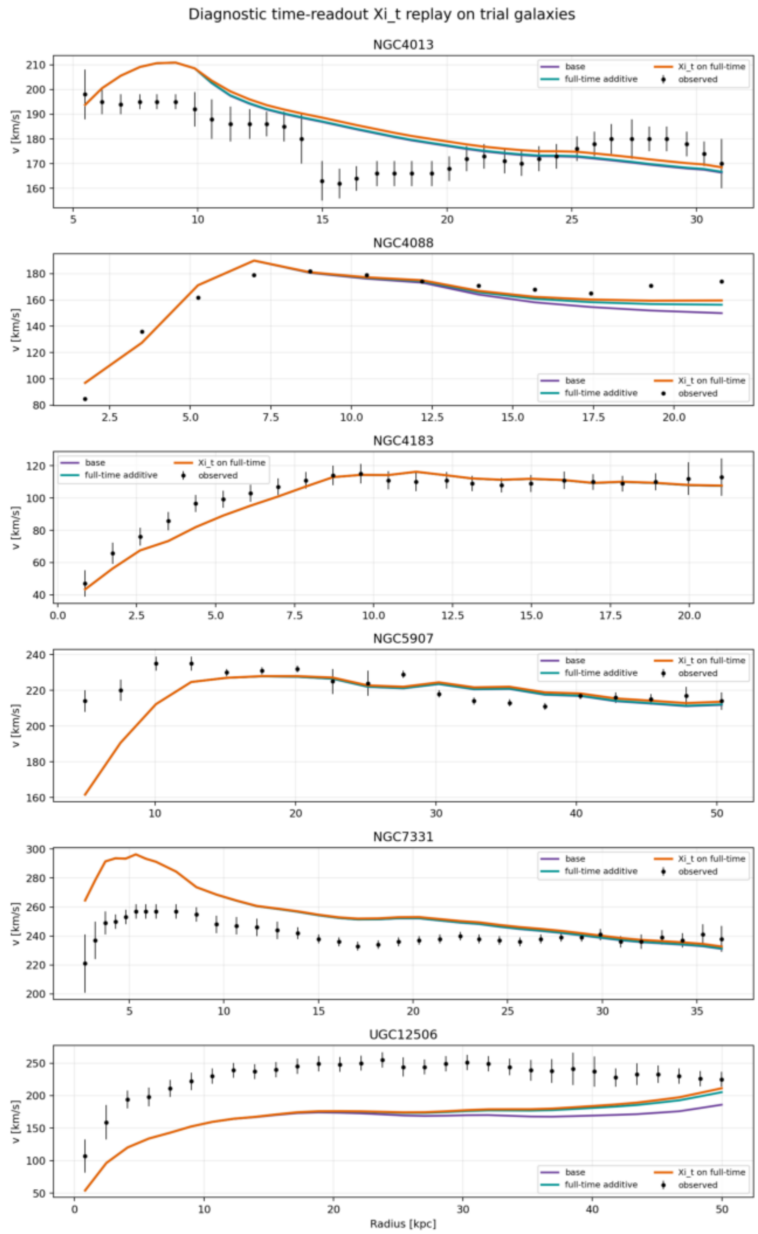


Figure 3: Diagnostic Ξ_t replay. The orange curve applies the small time-readout factor to the full-time morphology proxy. The effect improves NGC4088 and UGC12506, is effectively neutral in NGC4183, and worsens the already close NGC4013, NGC5907, and NGC7331 curves. This non-universal pattern is important: the time-readout factor is not acting as a generic amplitude rescue.

The Ξ_t diagnostic strengthens the claim boundary rather than weakening it. If the clock/readout factor were merely a hidden curve-fitting knob, it would be expected to improve all trial galaxies once turned on. Instead, it helps the two strongest stress cases in this sample: NGC4088, where warp/history/asymmetry is explicit, and UGC12506, where high-spin edge-on projection-history remains underpredicted. It is neutral or harmful for more regular or already saturated projection kernels. This is only diagnostic evidence, but it is consistent with the intended Tau Core interpretation: a time-readout term should be morphology/path-state dependent, not universal.

The replay was therefore converted into manifest-readiness and double-count audits rather than an endpoint claim, using the source-token priority protocol of Subsection 2.6. NGC4183 remains a weak/null control. NGC4013, NGC5907, and NGC7331 reject or fail to support the current Ξ_t proxy. The failure modes are instructive rather than generic negative results. NGC4013 worsens because the existing mixed warp/vertical-overlay kernel already encodes the source structure that the generic clock proxy tries to rescale; the added factor therefore double-counts the same geometry and moves an already close curve in the wrong direction. NGC7331 worsens more mildly because the active vertical/outer-warp readout is a broad-window proxy: the available phase information is already smeared into the mixed kernel, so the clock factor over-rescales a saturated component instead of adding a separately frozen clock/readout channel. NGC5907 is effectively neutral to slightly worse, consistent with an edge-on projection kernel that is already saturated at the present proxy level. Thus the priority protocol must allow $\Xi_t = 1$ when no independent clock/readout evidence exists; a degraded replay is a rejection of the current clock proxy, not a post-hoc invitation to retune it from the residual.

NGC4088 is the clearest example of this discipline. The clock-only replay improves the base projection curve, and the additive-plus-clock stress curve is numerically better than the accepted additive curve. However, a source-space overlap audit shows that the current Ξ_t ingredients—orientation mismatch, onset-side radial support, warp strength, and morphology-history phase—are already used by the additive warp/history kernel. The accepted combined route is therefore not the lower-RMSE additive-plus-clock stress curve. It is the caveated accepted additive warp/history endpoint with $\Xi_{\rm eff} = 1$, RMSE $= 11.62$ $\mathrm{km\,s^{-1}}$. The clock-only replay is preserved as a control; reopening a time-projection endpoint for NGC4088 would require new residual-blind clock/readout evidence that is demonstrably non-overlapping with the active additive morphology kernel under the T/A source-token priority test.

5 UGC12506 stress case: channel separation and source-derived closure

UGC12506 is kept separate from the general candidate audit because it plays a different role. It is the main stress case where the simpler projection and time/projection controls move in the right direction but still leave a large deficit. This section therefore records the source-review gates, the channel-separation ledger, and the source-derived closure candidate without promoting the galaxy to an endpoint success. UGC12506 is not used as a validation case in this manuscript. It is a stress-test and hypothesis-generation case: its role is to reveal which additional source-side readout channels would have to be frozen and transferred to independent galaxies before they can count as evidence.

5.1 Ξ_t source-review route

UGC12506 remains the active $P1$ source-review target for a future accepted $\Xi_t(R)$ manifest. It has strong high-inclination, extended-H I, asymmetric-envelope, low-density, and high-spin source context from the HIghMass VLA route. However, the foreground/path object term is not established, so the current source-only policy is to set that path term to zero unless a later cone or path-environment review supports it. Its remaining open step is a residual-blind high-spin/envelope clock proxy, not a foreground-rescue term.

As a first UGC12506 shell, the source-only clock proxy can be written

$$\Xi_t(R) = 1 + \epsilon_t K_t(R),$$

where K_t is assembled from high-spin settling, edge-on position–velocity clock-slice, diffuse H I envelope settling, and the caveated approaching/receding asymmetry component, while the path-environment component is set to zero. The current source-shell artifact gives $\epsilon_t \simeq 0.024$ from a bounded source-load rule. This is not an accepted endpoint: the remaining blocker is the accepted origin of the ϵ_t normalization law. The bounded shape of that rule is now itself separated from the open scale: given a nonnegative source load L, the minimal dimensionless, monotone, null-recovering, saturating response map is

$$\epsilon_t = \epsilon_{\rm cap}\frac{L}{1+L}.$$

For UGC12506 the present source load gives $L \simeq 2.14$, hence $L/(1+L) \simeq 0.68$. The open part is not this bounded shape, but the Tau-side or class-level origin of the small cap $\epsilon_{\rm cap}$. For the present source-shell replay we therefore freeze $\epsilon_{\rm cap} = 0.035$ only as a conservative small-mismatch protocol cap, not as a universal Tau Core constant. The cap is chosen inside the linearized time-readout regime: for $\Xi_t = 1 + \epsilon_t$, $\Xi_t^2 = 1 + 2\epsilon_t + \epsilon_t^2$, so the neglected quadratic correction is $\epsilon_t/2$ of the linear term. Requiring this ratio to be at most 2% gives $\epsilon_t \le 0.04$; the frozen cap 0.035 is below that bound, with a maximum quadratic-to-linear ratio 0.0175 and a maximum squared-velocity multiplier shift $(1.035)^2 - 1 \simeq 0.071$. Thus the UGC12506 shell now has a bounded source-load shape and a predeclared linear-regime cap, but it still does not become an accepted endpoint until the cap policy is promoted by an accepted-manifest or deeper Tau-side clock-geometry gate. The current accepted-manifest gate therefore returns `U12506_XI_T_ACCEPTED_MANIFEST_NOT_READY`: the shell passes the no-residual-leakage, zero-path-policy, bounded-shape, and cap-protocol checks, but remains blocked by the unreviewed $K_t(R)$ envelope mapping and the unfilled clock/readout settling proxy. This is the intended protocol outcome: UGC12506 is a strong source-shell target for a future Ξ_t endpoint, not an endpoint result in this draft. The corresponding source-review packet is now prepared with six review obligations and five allowed responses. It explicitly forbids using rotation residuals, endpoint RMSE, baseline ranks, wrong-family Tau scores, post-hoc cap changes, or foreground-rescue arguments to promote the route. Its status is `U12506_XI_T_SOURCE_REVIEW_PACKET_READY_RESPONSE_PENDING`; the next action is an independent source response, not endpoint scoring. The response-intake validator has also been exercised on a completed residual-blind reviewer response. The reviewer selected `ACCEPT_KT_CARRY_CAP_AS_CAVEATED_INTERVAL`: the high-spin, edge-on PV/envelope, and extended H I support route may be carried forward as a caveated interval/control manifest, while the asymmetry term remains a caveated phase component, the path term remains zero, and $\epsilon_{\rm cap} = 0.035$ remains a protocol cap rather than a universal Tau Core constant. The intake validator records the response as usable for the manifest gate but still endpoint-blocked. The accepted-manifest gate therefore returns a caveated-interval/control-ready, endpoint-blocked status; the exact machine-readable status is recorded in the released gate summary. Finally, a reviewer handoff has been generated for this route. It contains a fillable response form, six source-review tasks, five allowed route-level responses, forbidden-input guardrails, and input hashes. This handoff is the artifact to send to an external or otherwise separated residual-blind reviewer. A portable handoff bundle is also generated at `review_bundles/ugc12506_xi_t_external_review_bundle.zip`. The bundle contains the prompt, usage note, fillable response form, source-review packet, source evidence tables, forbidden-input ledger, and a relative-path hash manifest. It is a review-handoff artifact only: a completed response must still pass the intake validator and accepted-manifest gate before any endpoint score is allowed. The resulting caveated control manifest is stored in the released `data/derived` artifact `ugc12506_xi_t_caveated_interval_control_manifest.csv`. It freezes $\epsilon_t \in [0, 0.0238438]$, $\Xi_t \in [1, 1.02384]$, and a zero path term for any future control replay.

It does not authorize a standard endpoint score. Running that frozen manifest as a caveated control replay gives a small but not decisive movement: the lower edge of the interval has RMSE 77.54 $\mathrm{km\,s^{-1}}$, while the upper edge has RMSE 76.49 $\mathrm{km\,s^{-1}}$. The improvement is therefore about 1.06 $\mathrm{km\,s^{-1}}$, and no observed point is contained inside the narrow Ξ_t interval, even allowing measurement errors. This is scientifically useful precisely because it is not an all-purpose rescue term: the source-reviewed cap moves the curve in the expected direction but is too small to close the UGC12506 deficit by itself. Thus UGC12506 remains a caveated control case, not a positive endpoint.

5.2 Θ_{morph} and Ξ_t channel separation

With the refined Tau morphology-state terminology, the UGC12506 Θ_{morph} and Ξ_t routes can now be separated cleanly. The Θ_{morph} replay has the additive form

$$v_\theta^2(R) = v^2_{\mathrm{projection/history}}(R) + A_\theta K_\theta(R),$$

so it is a morphology-state phase or late-settling diagnostic. The Ξ_t route instead has the multiplicative interval-control form

$$\Xi_t(R) = 1 + \epsilon_t K_t(R),$$

so it is a clock/readout control. The released separation gate records a "channels separated, endpoint still blocked" status: the formula roles are distinct, but the high-spin/envelope/asymmetry source context still partially overlaps and the path term remains unestablished. Consequently the separation is an architectural clarification and a control ledger improvement, not an endpoint validation claim. The follow-up source-nonoverlap gate sharpens this further. It assigns the late-settling outer radial shape to the Θ_{morph} channel and the small-mismatch ϵ_t cap to the Ξ_t channel, but keeps high spin, the extended low-density H I envelope, edge-on position–velocity geometry, and approaching/receding H I asymmetry as shared or partially shared context. The gate therefore returns a partial-nonoverlap status: a combined-control replay is allowed with those assignments frozen, but a combined endpoint remains blocked until non-overlapping clock/path evidence is supplied. Running that ledger-frozen combined-control replay gives a useful but still non-endpoint result. The Θ_{morph}-only diagnostic has RMSE 64.12 $\mathrm{km\,s^{-1}}$. The ledger-strict combined control, which uses the Θ_{morph} late-settling shape plus only the Ξ_t small-mismatch cap, improves this to 60.31 $\mathrm{km\,s^{-1}}$. By contrast, the caveated shared-context $K_t(R)$ stress curve reaches only 63.21 $\mathrm{km\,s^{-1}}$. This is the desired claim-safe behavior: the cleaner ledger-separated control moves in the helpful direction, while the shared-context route is preserved as a caveated stress test rather than being used as an endpoint rescue.

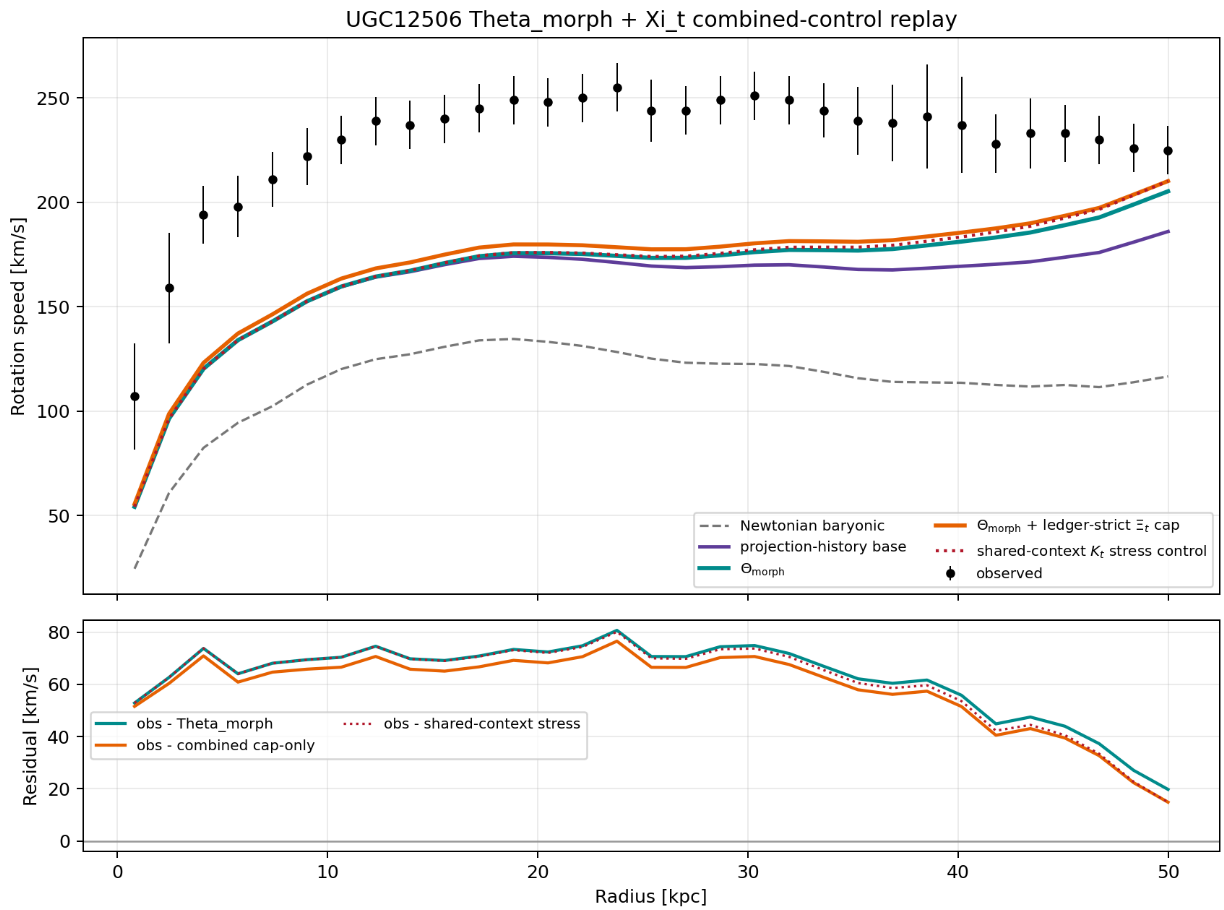


Figure 4: UGC12506 $\Theta_{\mathrm{morph}} + \Xi_t$ combined-control replay. The orange curve is the ledger-strict control: the additive Θ_{morph} phase curve multiplied only by the Ξ_t protocol cap. The dotted red curve is the shared-context $K_t(R)$ stress control and remains caveated because its source evidence overlaps the morphology channel. This figure is a control replay, not endpoint validation.

Figure 5 shows the same result in direct comparison with the Newtonian baryonic, source-native NFW-HSE, and projection-history curves. The residual panel is included to make clear that the combined-control curve reduces the systematic underprediction but does not close the UGC12506 gap. The comparison is therefore evidence for channel accounting and kernel development, not a population or endpoint validation.

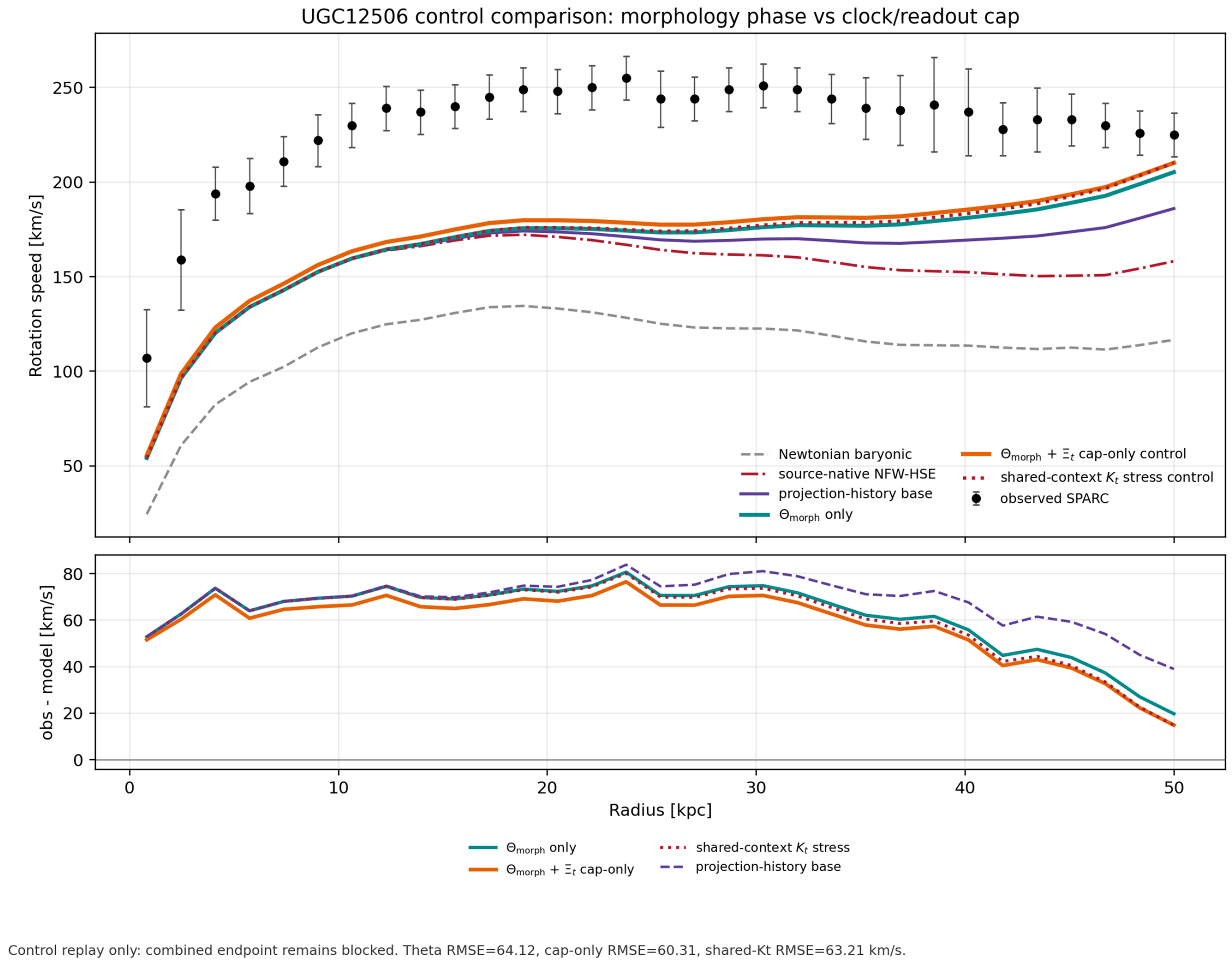


Figure 5: UGC12506 comparison rotation curves. The ledger-strict $\Theta_{\text{morph}} + \Xi_t$ cap-only control is the best current source-ledger control in this comparison, but all displayed Tau Core curves remain below the observed rotation curve over much of the disk. The shared-context K_t stress curve is shown separately to avoid hiding the source-overlap caveat.

6 Cross-galaxy controls for the time/projection layer

6.1 Control-galaxy audit of the current time/projection layer

We also asked whether the same time/projection–morphology layer improves the other inspected control galaxies. This is again a diagnostic audit rather than endpoint validation. The answer is selective. Relative to the morphology-only proxy, the combined time/projection layer improves clearly for UGC12506 (ΔRMSE $= -3.81$ km s^{-1}) and NGC4088 (-1.04 km s^{-1}), is nearly neutral for NGC4183 (-0.01 km s^{-1}) and NGC5907 ($+0.03$ km s^{-1}), and worsens the current proxy for NGC7331 ($+0.16$ km s^{-1}) and NGC4013 ($+0.53$ km s^{-1}). This behavior is important for the claim boundary. The time/projection layer is not acting as a universal amplitude knob; it helps primarily in cases where the source context indicates strong history, asymmetry, or projection-clock relevance. Where it worsens, the cause is not treated as a free refit opportunity. In NGC4013 the current time/projection factor overlaps the already active warp/vertical-overlay mixed kernel, so it double-counts rather than supplies an independent clock/readout channel. In NGC7331 the vertical/outer-warp window is broad enough that the mixed kernel already carries the available phase information; the extra factor therefore over-rescales a saturated proxy. These rows are retained as negative-control evidence for the default rule $\Xi_t = 1$ unless independent, non-overlapping clock/readout evidence is source-frozen before scoring.

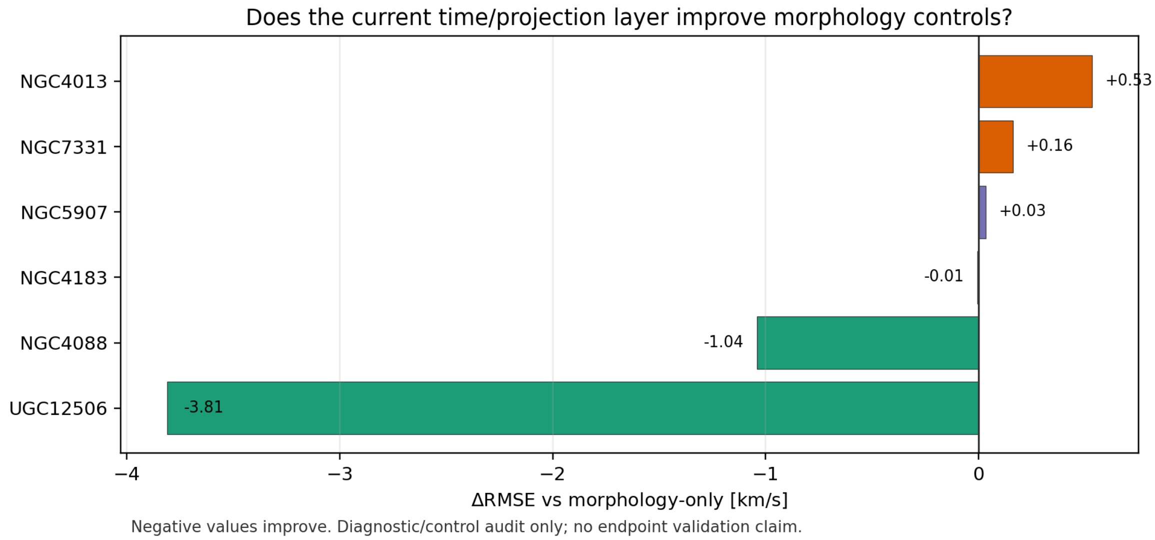


Figure 6: Control-galaxy audit of the current time/projection–morphology layer. Negative values improve relative to the morphology-only proxy. The effect is selective: it improves clearly for UGC12506 and NGC4088, is nearly neutral for NGC4183 and NGC5907, and worsens the current proxy for NGC7331 and NGC4013. This is a diagnostic/control audit, not endpoint validation.

6.2 Why the improving cases improve

The two clearly improving rows in Fig. 6 are also the two rows for which the source-side morphology makes a time/projection readout contribution most plausible. This does not prove a final Tau Core time-projection law, but it explains why the current diagnostic layer improves these cases rather than behaving as a universal gain term.

For UGC12506, the source context is a high-inclination, high-spin system with an extended H I envelope and strong projection-history stress. The simpler Tau morphology and projection-history curves underpredict the measured rotation over much of the disk. The ledger-strict $\Theta_{\mathrm{morph}} + \Xi_t$ cap-only control improves the Θ_{morph}-only diagnostic from 64.12 to 60.31 $\mathrm{km\,s^{-1}}$. The physical reading is that the late-settling outer radial shape and the small clock/readout interval cap both move the curve in the expected direction for an edge-on high-spin envelope system. The source-overlap ledger remains essential: the improvement is evidence for a projection/time-relevant channel, not an accepted endpoint.

For NGC4088, the source context is qualitatively different. The system is a strong warp/history/asymmetry case, so the additional trajectory/phase layer is expected to matter more than in regular disks. The full-time morphology proxy improves the base projection curve from 11.62 to 9.39 $\mathrm{km\,s^{-1}}$, and the diagnostic Ξ_t replay reaches 8.35 $\mathrm{km\,s^{-1}}$. The interpretation is therefore not simply "more amplitude". The added layer is acting where the source morphology already indicates unsettled/asymmetric readout structure. Because the clock ingredients overlap the accepted additive warp/history kernel, the lower-RMSE clock curve is still retained as a control rather than promoted to the accepted endpoint route.

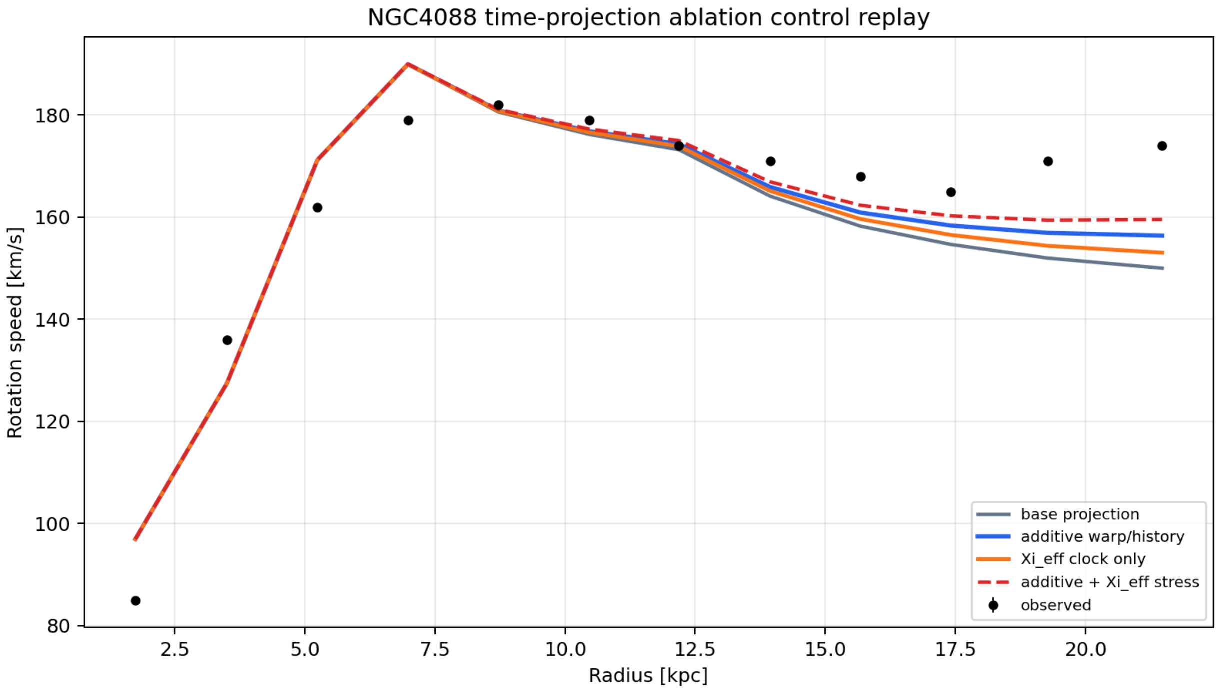


Figure 7: NGC4088 time/projection ablation control. The improvement is largest when the trajectory/phase proxy and the small Ξ_t replay are both active, which is consistent with the source-side warp/history/asymmetry classification. The figure is a control replay: it motivates the time/projection channel, but does not promote a final clock endpoint because the clock evidence is not yet independent of the accepted additive warp/history kernel.

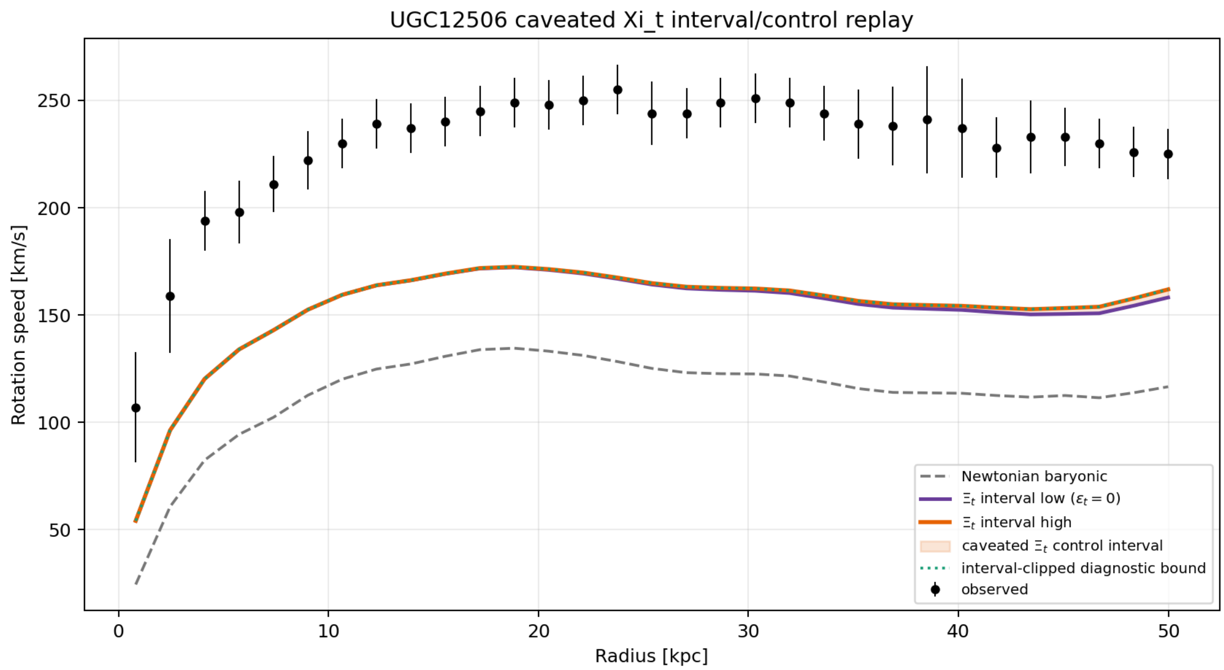


Figure 8: UGC12506 caveated Ξ_t interval/control replay. The shaded band is the source-reviewed control interval allowed by the completed reviewer response. The interval improves the source-native base slightly but does not contain the observed rotation curve. The dotted clipped curve is a diagnostic lower bound inside the frozen interval, not a fitted model.

7 UGC12506 source-native mass/envelope–closure route

7.1 Source-native ablation and post-diagnostic beta closure

The next-gate ledger identifies UGC12506 as the case where the remaining deficit is least plausibly solved by a generic clock multiplier alone. We therefore ran the first source-native mass/envelope–closure ablation using the published NFW/high-spin-envelope source route. The kernel and amplitude are frozen before scoring from source-side quantities; the observed rotation curve enters only in the replay score. The result improves the older R_d-proxy NFW/HSE branch slightly, from 77.86 to 77.54 $\mathrm{km\,s^{-1}}$ RMSE, and improves strongly over the pure edge-on/envelope/asymmetry and source-envelope controls, both near 102.45 $\mathrm{km\,s^{-1}}$. It also improves over the baryonic carrier (116.02 $\mathrm{km\,s^{-1}}$). However, it remains far worse than the prior diagnostic Tau/MOND/TPG-level references, whose best diagnostic RMSE is 37.36 $\mathrm{km\,s^{-1}}$. This is therefore useful source-side evidence for a real mass/envelope–closure channel, not an endpoint validation and not a closed UGC12506 solution.

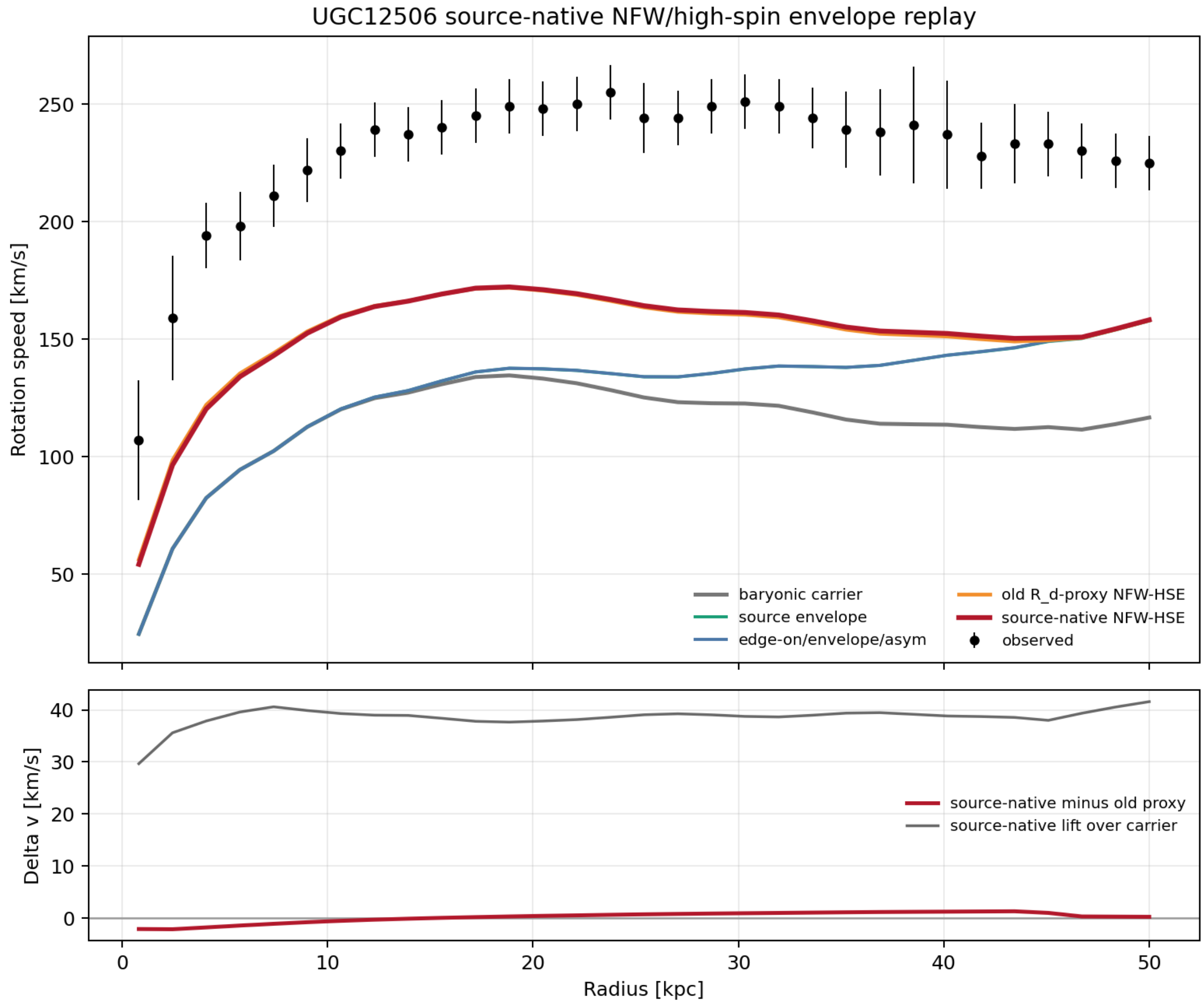


Figure 9: UGC12506 source-native NFW/high-spin-envelope replay. The red curve uses source-native NFW/HSE parameters and improves the older R_d-proxy branch and lower source-envelope controls, but it still underpredicts the observed curve substantially. The replay is frozen before scoring and remains a control result, not an endpoint validation.

The shape failure is therefore mostly an amplitude-normalization failure. A separate diagnostic fit keeps the NFW/HSE shape fixed and allows only one velocity-squared amplitude multiplier. The residual-aware optimum is $\beta_{\rm diag} = 3.876$, which reduces the RMSE to 7.48 $\mathrm{km\,s^{-1}}$. This diagnostic is not endpoint evidence, because $\beta_{\rm diag}$ uses the observed curve. It is nevertheless useful: it says that the source-native NFW/HSE shape is already close enough, and that the missing object is a source-side normalization law.

A first source-only candidate for that normalization is

$$\beta_{\rm cl} = 1 + \frac{\lambda_{\rm spin}}{\lambda_{\rm ref}} \left(\frac{\chi^2_{\rm iso}}{\chi^2_{\rm NFW}} - 1 \right)_+ + \sin^2 i \, \max\!\left(\frac{i - 80^\circ}{10^\circ}, 0 \right),$$

with $\lambda_{\rm ref} = 0.10$. The terms are dimensionless: high spin amplifies the source-native NFW closure preference, while the last term is the edge-on projection load. For UGC12506 this gives $\beta_{\rm cl} =$

3.954, within 2.1% of the residual-aware diagnostic normalization, and gives 7.67 $\mathrm{km\,s^{-1}}$ RMSE. The calculation uses only source-frozen spin, Table-5 halo preference, and inclination, but the formula was written after the normalization diagnostic. It is therefore a post-diagnostic source-candidate replay, not a source-frozen endpoint law. A future paper-grade promotion would require predeclaring this $\beta_{\rm cl}$ rule and testing it on independent high-spin edge-on systems before endpoint scoring. This formula is not counted as evidence on UGC12506. Its only admissible role in the present paper is to define a frozen transfer hypothesis for independent high-spin edge-on systems.

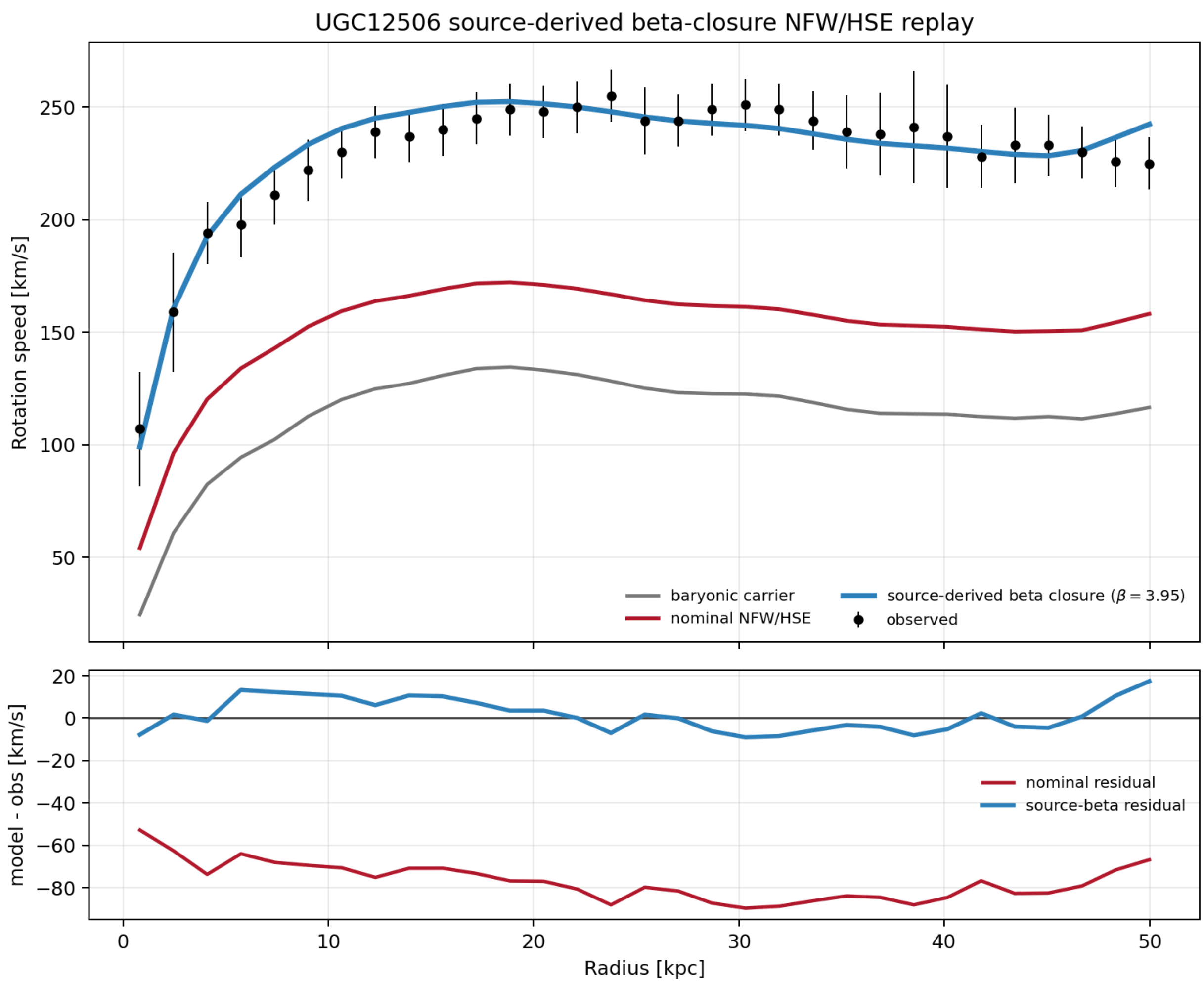
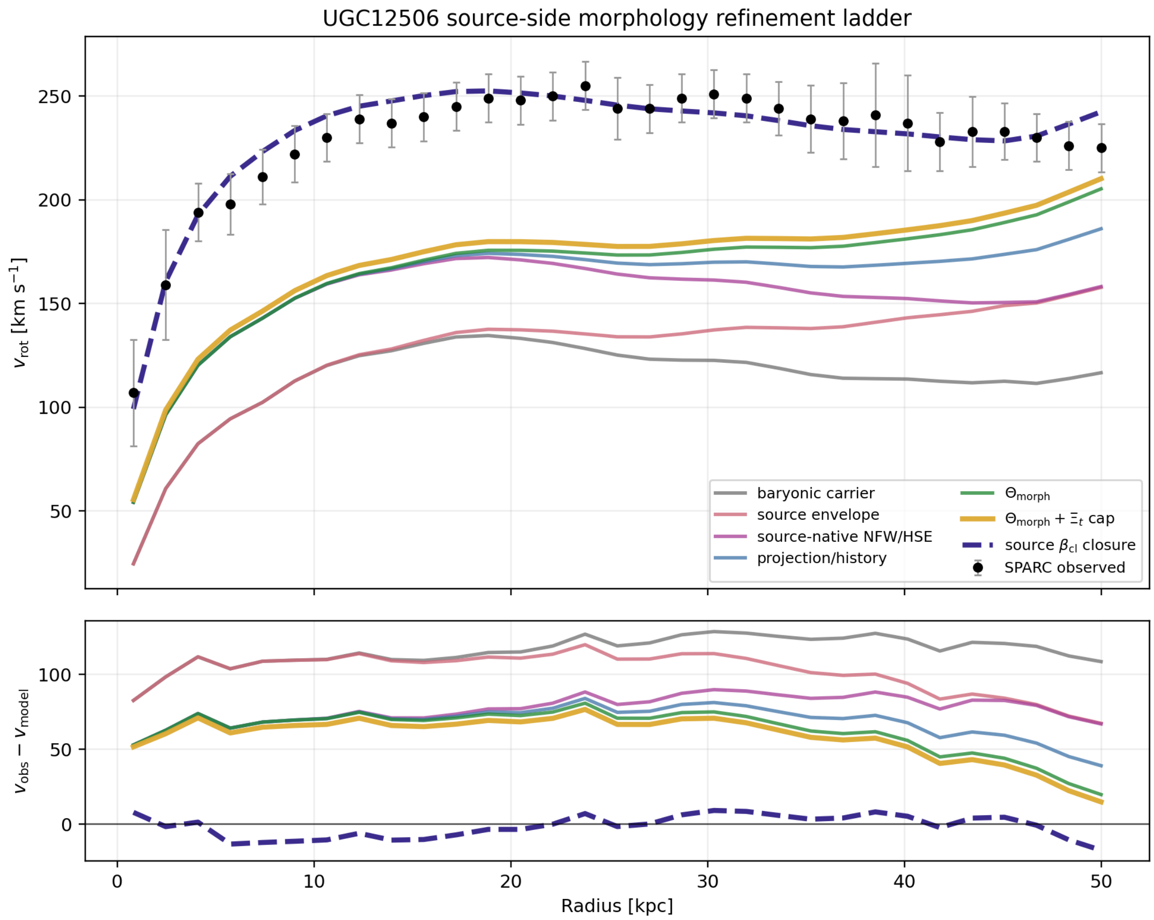


Figure 10: UGC12506 source-derived $\beta_{\rm cl}$ closure replay. The blue curve uses a source-only beta factor from spin, source-native NFW preference, and edge-on projection load. It nearly reproduces the residual-aware normalization scale and follows the observed curve closely, but remains a post-diagnostic candidate until transferred to independent source-frozen cases.

The UGC12506 sequence is therefore a second morphology-refinement example, but with a stronger caveat than the NGC0891/NGC4217 edge-on vertical/halo controls. Here the first six rows are source-frozen or ledger-strict control replays, while the final $\beta_{\rm cl}$ closure row is a post-diagnostic source candidate. The scientific use of the sequence is not to claim that UGC12506 has been solved. It is to show what a source-side refinement ladder looks like when increasingly specific Tau morphology-state channels are activated: baryonic carrier, envelope support, source-native mass/envelope closure, observer/projection history, morphology-state phase, and finally a clock/readout cap.

Step	Readout curve	RMSE	Status
1	baryonic carrier	116.02	baseline/control carrier
2	source envelope support	102.48	source-frozen envelope branch
3	source-native NFW/HSE	77.54	source-native mass/envelope closure control
4	projection/history increment	69.18	observer/projection-history control
5	$\Theta_{\rm morph}$	64.12	Tau morphology-state phase diagnostic
6	$\Theta_{\rm morph} + \Xi_t$ cap	60.31	ledger-strict combined control
7	source $\beta_{\rm cl}$ closure	7.67	post-diagnostic source candidate; not endpoint validation

Table 8: UGC12506 morphology-refinement ladder. Values are RMSE in $\mathrm{km\,s^{-1}}$. The ordered improvement is informative because each pre-$\beta_{\rm cl}$ step corresponds to an explicitly recorded source-side channel or control artifact. The final source $\beta_{\rm cl}$ row is shown to indicate the remaining amplitude/closure route, but it is not promoted to endpoint evidence in this manuscript.



Ladder uses pre-existing source/control artifacts. $\beta_{\rm cl}$ is post-diagnostic candidate, not endpoint validation.

Figure 11: UGC12506 source-side morphology refinement ladder. The curves move progressively toward the observed rotation data as more specific source-side readout channels are activated. This is not curve fitting: the steps are assembled from pre-existing source/control artifacts and the residuals are used only for replay scoring. The dashed $\beta_{\rm cl}$ curve is the post-diagnostic source-candidate closure route and is not counted as endpoint validation.

To prevent this encouraging result from becoming a same-galaxy rescue, we also freeze an independent transfer predeclaration. Using only SPARC source proxies for inclination, $R_{\rm HI}/R_d$, H I mass, $V_{\rm flat}$, and quality flag, the transfer gate finds 11 non-UGC12506 candidates and authorizes no replay score yet. The top source-acquisition targets are UGC11455, ESO563-G021, IC4202, NGC0891, NGC4013, NGC2841, NGC4157, and NGC4217. Each candidate must obtain source-native $\lambda_{\rm spin}$, $\chi^2_{\rm NFW}$, $\chi^2_{\rm ISO}$, halo-fit provenance, and PV/envelope notes before the $\beta_{\rm cl}$ rule can be replayed as an independent transfer test. As a first blocker-reduction step, the Li et al. SPARC halo-fit catalogue [10] was used through VizieR to fill the pISO and NFW reduced-χ^2 fields for all 11 predeclared candidates. This does not unlock replay, because $\lambda_{\rm spin}$ and PV/envelope evidence remain unfrozen, but it changes the practical transfer queue: the highest edge-on/massive-H I proxies are not necessarily UGC12506-like NFW-preference objects. In the current catalogue readout, positive NFW-preference load appears most clearly for NGC0891, NGC7331, NGC2841, NGC0801, and weakly NGC4013. The resulting priority gate therefore keeps NGC0891 and NGC7331 as the first source-review targets for this specific $\beta_{\rm cl}$ transfer route, retains NGC2841, NGC0801, and NGC4013 as weak/secondary targets, and preserves the pISO-preferred systems as controls or alternative-branch candidates rather than forcing the UGC12506 normalization rule onto them. The first source-freeze preflight for these two primary targets accepts PV/envelope context for both objects, but freezes no direct $\lambda_{\rm spin}$ value. Thus no $\beta_{\rm cl}$ replay is authorized: the next admissible step is either a direct source-native spin acquisition or a predeclared source-only spin-proxy rule before any transfer residual is inspected. A direct-source search only partially reduces this blocker. NGC7331 has a published disc-spin-like value, $\lambda = 0.423$, in the lognormal self-gravitating disc analysis of Marr [12], but that quantity is not the same as the halo/envelope $\lambda_{\rm spin}$ slot used in the $\beta_{\rm cl}$ rule. NGC0891 has no accepted direct source-native $\lambda_{\rm spin}$ value in the current cache; the available NGC891-like spin context is model-analogue material only. Therefore no direct lambda source currently authorizes replay. A follow-up NGC0891 source hunt strengthens the physical transfer motivation but not the missing normalization slot. Deep H I and extraplanar-gas studies show a massive lagging halo, low-angular-momentum accretion context, vertical rotation gradients, and projection-sensitive halo modelling [15, 5, 6, 11]. These are exactly the kind of source-side facts that make NGC0891 a relevant halo/envelope transfer target, but none of them supplies the dimensionless halo/envelope $\lambda_{\rm spin}$ required by the $\beta_{\rm cl}$ slot. Thus the source hunt changes the interpretation from “weak target” to “well-motivated but still normalization-blocked target”, not to a replay. The explicit definition-conversion gate preserves this as a negative control: the Marr value is accepted as source-side angular-momentum context, but direct substitution is rejected. If one inserted $\lambda = 0.423$ into the $\beta_{\rm cl}$ slot without conversion, the NGC7331 beta factor would be 1.73, compared with 1.23 for the declared source-proxy candidate. The larger number is not a problem by itself; the problem is that no residual-blind disc-to-halo/envelope conversion functional has been accepted. As a conservative control, we also compute a Bullock-like disk-inferred spin conversion proxy rather than relying only on the source-declared exposure proxy. Following the standard disk/halo angular-momentum normalization logic [13, 2], we form

$$j_{\rm disk} = 2R_d V_{\rm flat}, \qquad R_{200} = \frac{V_{200}}{10H_0}, \qquad \lambda'_{\rm disk} = \frac{j_{\rm disk}}{\sqrt{2}R_{200}V_{200}},$$

using only source-side SPARC disk quantities and the Li et al. NFW-flat V_{200} field. This gate is dimensionless and residual-blind, but it is not accepted as the Tau-side $\beta_{\rm cl}$ halo/envelope spin slot. It gives $\lambda'_{\rm disk} = 0.035$ for NGC0891 and 0.033 for NGC7331, substantially below the corresponding source-declared exposure-proxy values 0.149 and 0.136. The disagreement is scientifically useful: it means the replay cannot be reopened by a hidden amplitude choice. An independent review must choose direct spin acquisition, the exposure proxy, the Bullock-like conversion proxy, or rejection of the transfer route before any $\beta_{\rm cl}$ replay. We implement the second option only as a blocker-reduction gate. The declared proxy uses bounded source loads from $R_{\rm HI}/R_d$, $V_{\rm flat}$, H I mass, and inclination,

$$\lambda_{\rm spin,proxy} = 0.10\,[1 + 0.35L_{\rm extent} + 0.25L_V + 0.25L_{\rm gas} + 0.15L_{\rm edge}]\,,$$

and combines it with the already frozen NFW-preference load only to prioritize independent source review. This gate selects NGC0891 as the only primary proxy-transfer review target and keeps NGC7331, NGC2841, NGC0801, and NGC4013 as secondary targets. It is not accepted as a direct $\lambda_{\rm spin}$ measurement and still authorizes no replay or endpoint score. For reproducibility and leakage prevention, this proxy route is packaged as an independent review bundle rather than silently promoted. The bundle contains the declared proxy fields, transfer queue, direct-source audit, definition-conversion rejection, the Bullock-like conversion-control files, forbidden inputs, review obligations, and a blank response form. Its current status is `READY_RESPONSE_PENDING`: the reviewer may accept the source fields, choose the exposure proxy, choose the Bullock-like disk-conversion proxy, require direct source-native spin, require a new residual-blind rule, keep the Marr NGC7331 value as context only or supply a genuine conversion rule, and restrict the transfer target set. The bundle itself still sets `beta_cl_replay_allowed=false` and `endpoint_scores_allowed=false`. Thus the proxy is reviewable without residual leakage, but not yet an accepted Tau-side amplitude law. A companion response-intake validator is also present. With no completed external response currently supplied, it records `RESPONSE_PENDING_ENDPOINT_BLOCKED`, with `selected_spin_normalization_route` still pending. Until such a route is independently accepted, proxy promotion, $\beta_{\rm cl}$ replay preflight, and endpoint scoring all remain disabled. The downstream spin-route prefreeze gate is already implemented and currently records `SPIN_ROUTE_PREFREEZE_BLOCKED_REVIEW_ROUTE_PENDING`, emitting no transfer values. Thus the paper has a concrete replay lock rather than an informal caveat. We also expose this lock as a four-row route-decision matrix. The available decisions are the source-declared exposure proxy, the Bullock-like disk conversion proxy, a future direct source-native spin measurement, or rejection of the route. The first two are computable but reviewable-not-accepted; the third is scientifically preferred but currently missing; the fourth is a valid route-level negative outcome. None of the four rows authorizes endpoint scoring. Finally, the beta-closure scoring launch gate is already present. It finds that all nine required launch inputs exist, but the review-route acceptance, prefreeze-value, and carrier-freeze gates are still blocked. The protocol skeleton therefore identifies the future scoring script as the only step allowed to read $v_{\rm obs}$, and keeps that step inactive until a source-frozen formula manifest exists. The additional carrier-freeze gate is needed because $\beta_{\rm cl}$ is an amplitude/closure factor, not a complete rotation-curve formula by itself. It records three possible carrier routes: a reviewable but not yet accepted baryonic-fast-packet stress carrier, a Li et al. NFW-fit diagnostic/control carrier, and the preferred but currently missing source-native NFW/HSE transfer carrier. The Li et al. route is useful for context and controls, but it is not endpoint-safe by default because its halo parameters are published rotation-curve fit products. This blocker is now converted into an explicit review packet rather than left as an informal caveat. The carrier review bundle contains the route decision matrix, source artifacts, forbidden-input ledger, and a blank response form. The response intake currently records `CARRIER_REVIEW_RESPONSE_PENDING_ENDPOINT_BLOCKED`. A future independent response may accept the minimal baryonic stress carrier, require a source-native transfer carrier, or reject the route, but it may not use endpoint residuals or baseline ranks to choose the carrier. We now also instantiate that intermediate formula-freeze gate explicitly. The gate writes the transfer formula manifest path with headers and zero rows, passes the halo-fit and priority-field checks, but blocks on independent route-review acceptance, source-frozen spin-route values, and an accepted carrier. Its status is `TRANSFER_FORMULA_FREEZE_BLOCKED_PREFREEZE_PENDING`. In other words, the formula shell

$$\beta_{\rm cl} = 1 + \frac{\lambda_{\rm spin}}{0.10} L_{\rm NFW} + L_{\rm edge}$$

is present as a declared transfer form, but no galaxy row is frozen into the manifest until the spin-normalization route is independently accepted. This gate reads no observed rotation curve and produces no score. The separate transfer scoring runner has also been added; in the current state it exits with `TRANSFER_SCORING_BLOCKED_LAUNCH_GATE`, writes an empty score table, sees the existing but empty formula manifest, and reads no observed rotation curve. The dormant control-scoring branch is implemented for a future accepted `BARYONIC_050_FAST_PACKET` carrier:

after the launch and formula-freeze gates pass, it computes

$$v_{\mathrm{readout}}(R) = \sqrt{\beta_{\mathrm{cl}}\, v^2_{\mathrm{baryon},0.5}(R)}$$

and writes both score-level and point-level artifacts. This keeps the score-producing stage technically present but leakage-locked. Finally, a no-v_{obs} scoring-contract dry run is now generated. It does not accept either review route and does not score. Instead, it asks whether the already implemented spin routes and the reviewable baryonic stress carrier would have enough non-observed inputs to execute if later accepted by independent review. The current dry run reports two contract-ready scenarios: `EXPOSURE_PROXY` and `BULLOCK_DISK_CONVERSION`, both paired with `BARYONIC_050_FAST_PACKET`. It writes 22 dry-run manifest rows and 656 prediction rows without reading observed velocities or residuals. Thus the remaining blocker is review/freeze authorization, not missing scoring infrastructure. The final pre-scoring handoff is an unlock packet with exact active-response requirements for the spin-route and carrier reviews. It includes example-only CSV rows for the two spin-route choices and the baryonic stress carrier, but it does not create active response files. Scoring remains blocked until an independent reviewer supplies those active files and the standard intake, prefreeze, formula-freeze, and launch gates pass. A post-review scoring launcher now runs that entire chain as one command. In the present package all chain scripts return successfully, but the launcher keeps the scoring runner blocked because zero active reviewer response files are present. Thus the scoring path is mechanically ready, while the scientific authorization remains external-review gated. To avoid accidental promotion of example rows, active reviewer responses are installed only through a validator that checks the incoming CSV schemas, rejects `EXAMPLE_ONLY` or pending rows, rejects endpoint/residual flags, and copies the files into the active paths only if both responses pass. In the present package the installer reports zero incoming active responses and therefore installs nothing. The current release also writes a scoring-readiness dashboard that condenses the dry-run contract, unlock packet, active-response installer, launcher, launch gate, formula-freeze gate, and dormant scoring runner into a single ledger. That ledger reports the present status as `SCORING_READINESS_BLOCKED_ACTIVE_RESPONSES_PENDING`: the no-v_{obs} contract and unlock packet are ready, but no score is authorized and no observed rotation curve is read until the two active review responses are supplied and validated. Thus the present result is a strong post-diagnostic derivation candidate, while the validation path is explicitly shifted to independent source acquisition.

8 Source-blind morphology refinement controls

8.1 Beta-transfer lock-type review and refinement ladder

The beta-transfer work also gives a useful example of how morphology refinement should be carried out without becoming curve fitting. The refinement is not allowed to choose a curve because the curve looks better. Instead, the only admissible sequence is:

$$\text{source morphology review} \;\rightarrow\; \text{frozen readout family} \;\rightarrow\; \text{control replay} \;\rightarrow\; \text{score.}$$

The replay score may trigger a review, but it may not decide which morphology-family replacement is used. The source-side replacement must be justified from independent observations such as inclination, dust lanes, extra-planar gas, radio/halo emission, H I support, or published projection/vertical-structure studies. This section is not evidence for beta closure. It is a methodological control showing how failed transfer branches are handled without residual-driven refitting:

the failed pure branch is preserved, and a replacement branch is allowed only after source-side morphology review.

The current beta-transfer control illustrates the full ladder. The global closure-transfer branch

$$v_\beta^2(R) = \beta_{\rm cl}\, v_{\rm carrier}^2(R)$$

is a deliberately blunt control: it asks whether the UGC12506-inspired closure normalization can be transported as a whole. For NGC0891 and NGC4217 this pure branch fails by overshooting. That failure is preserved as a negative branch-level control, not hidden. The residual-blind source review then asks a different question: are these galaxies actually pure beta/compact-transfer cases, or are they edge-on vertical/halo systems whose readout should be spatially gated? The source ledger rejects the pure beta/compact primary lock for both rows and supports an edge-on vertical/dust/H I/radio-halo mixed interpretation, using independent edge-on dust, H I halo, extra-planar gas, and radio/halo evidence [14, 9, 15, 4, 8]. The exact review artifacts are `data/derived/beta_transfer_source_side_morphology_review_gate.csv` and `data/derived/beta_transfer_lock_type_source_review.csv`.

The first source-locked refinement is therefore not a new amplitude fit but an edge-on vertical/halo gate:

$$v_{\rm EVH}^2(R) = v_{\rm carrier}^2(R)\left[1 + (\beta_{\rm cl} - 1)K_{\rm EVH}(R)\right],$$

where

$$K_{\rm EVH}(R) = {\rm smoothstep}\left(\frac{R - R_{\rm on}}{R_{\rm full} - R_{\rm on}}\right), \qquad R_{\rm on} = 2R_s, \quad R_{\rm full} = \min(3R_s, R_{\rm max}).$$

This version says that the transferred closure load should become active only where the source-side edge-on vertical/halo component is expected to project strongly into the rotation readout. It improves both galaxies relative to the carrier and avoids the pure global-beta overshoot.

The second refinement follows a further source-side observation: the vertical dust/halo and extra-planar gas evidence for these edge-on systems is not only an extreme outer-tail feature. It is distributed over the disk/halo interface. That motivates a distributed vertical/halo gate,

$$v_{\rm DVH}^2(R) = v_{\rm carrier}^2(R)\left[1 + (\beta_{\rm cl} - 1)K_{\rm DVH}(R)\right],$$

with

$$K_{\rm DVH}(R) = {\rm smoothstep}\left(\frac{R - R_s}{2R_s - R_s}\right).$$

Again, the radial window is not chosen by minimizing the rotation-curve residual. It is the predeclared translation of the source claim that the vertical/halo component is distributed through the disk-halo interface rather than activated only at the far edge.

Galaxy	Carrier	Pure β	EVH	DVH	Source-side interpretation
NGC0891	44.15	74.35	30.20	23.07	Pure global beta overshoots; edge-on dust/H I halo and extra-planar gas support vertical/halo gating, with distributed activation preferred by the source morphology.
NGC4217	47.62	55.00	37.66	27.23	Pure global beta remains too blunt; edge-on vertical/radio-halo context supports a mixed vertical/halo readout, with distributed activation improving the replay.

Table 9: Morphology-refinement ladder for the beta-transfer control rows. Entries are RMSE values in $\mathrm{km\,s^{-1}}$. The table is a control replay, not population validation. The important feature is the ordered improvement: source review rejects the pure branch, freezes a more specific morphology family, and only then scores the replay.

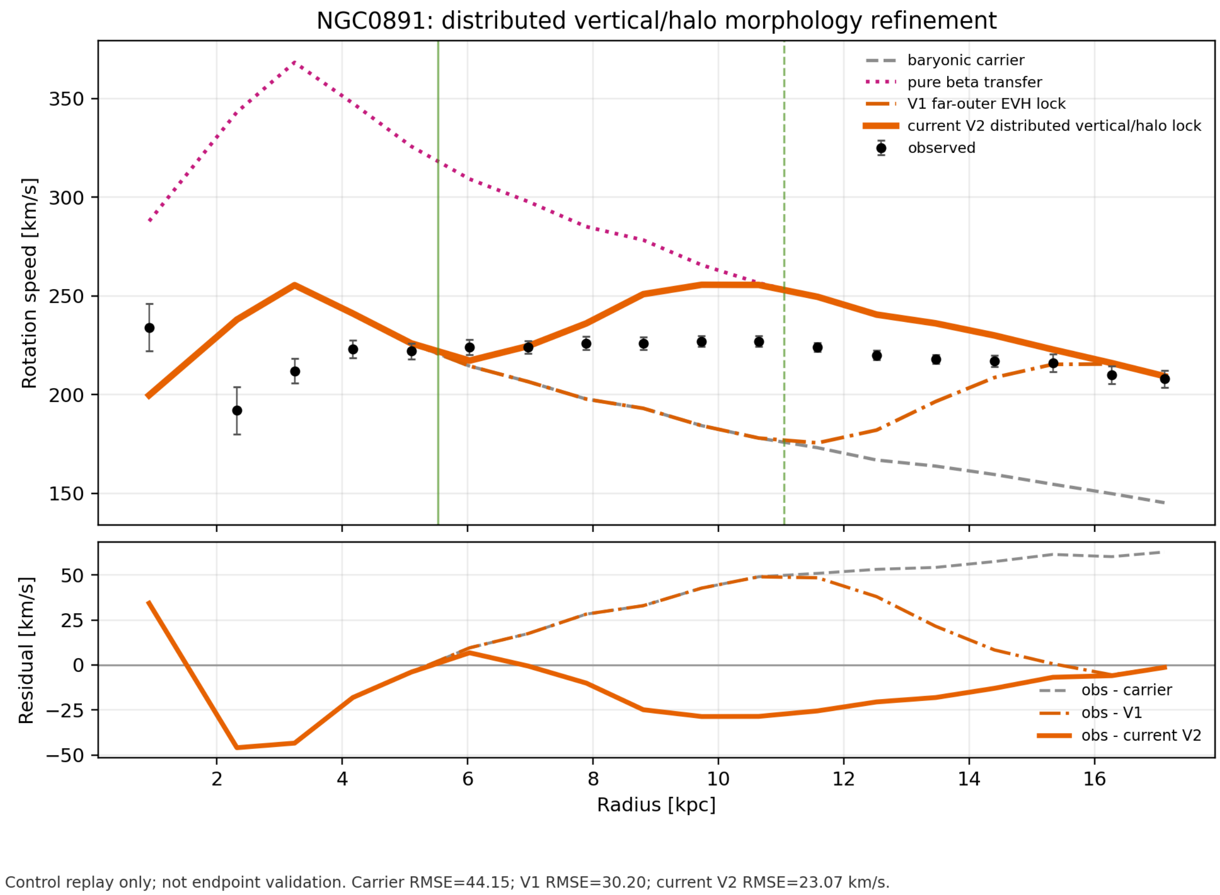


Figure 12: NGC0891 morphology-refinement replay. The pure beta-transfer branch overshoots, while the source-frozen edge-on vertical/halo and distributed vertical/halo kernels progressively move the predicted curve toward the measured rotation data. The refinement is source-side: the lock-type review uses edge-on dust, extra-planar gas, and H I halo evidence, not the residual shape, to replace the pure branch.

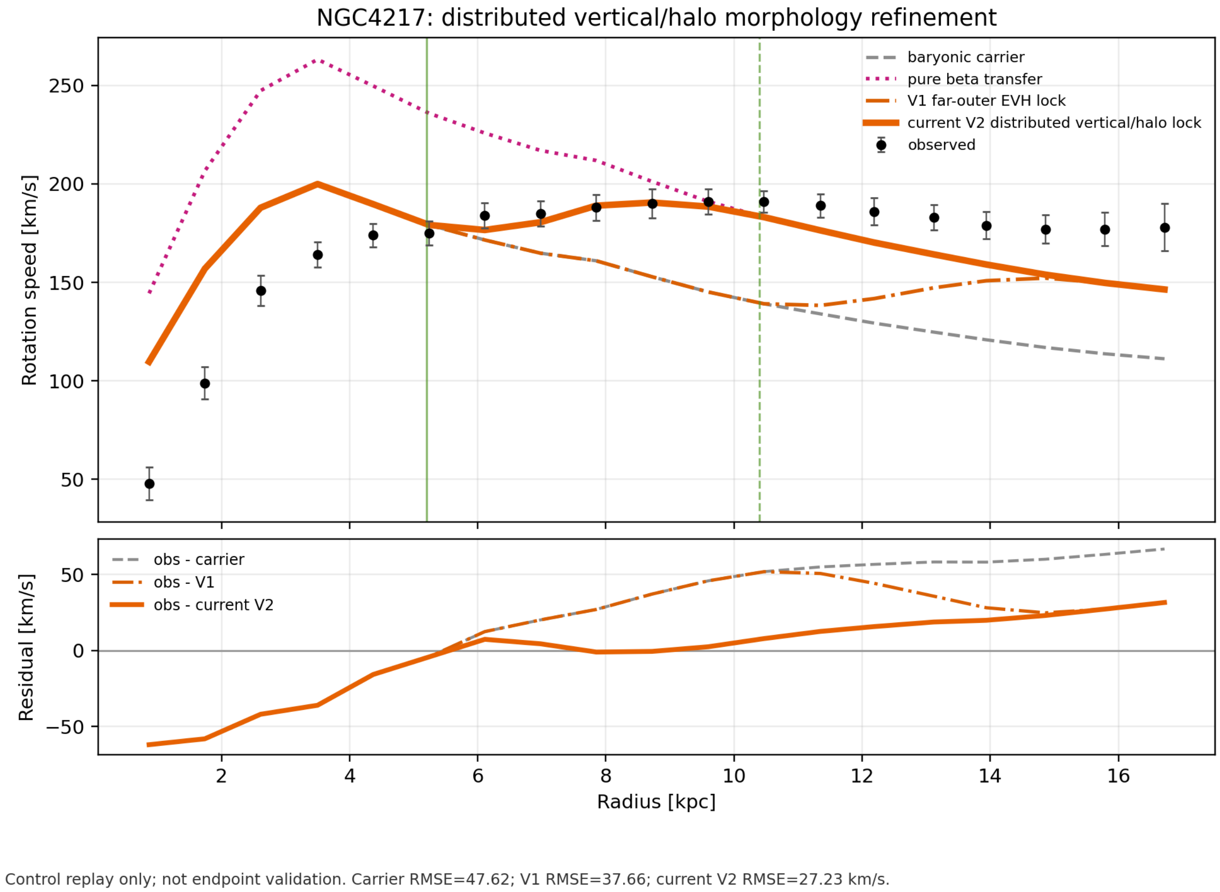


Figure 13: NGC4217 morphology-refinement replay. As in NGC0891, the curve improves when the pure closure-transfer branch is replaced by a source-supported edge-on vertical/halo readout, and improves further when the vertical/halo contribution is treated as distributed through the disk-halo interface. This is a morphology-refinement control, not an endpoint claim.

This ladder is important for the interpretation of failures. A bad replay is not automatically a Tau Core failure, and it is not automatically an invitation to refit. It is a source-review trigger. If independent observations support a more precise readout-relevant morphology, a replacement family may be frozen and replayed. If they do not, the failed branch remains a negative control. In this sense the paper tests a stronger operational claim than “can a curve be made to fit”: it tests whether progressively more source-complete, residual-blind morphology descriptions select kernels whose rotation readouts follow the measured curves more closely.

8.2 Additional refinement patterns: positive, guardrail, and caveated cases

The same source-blind refinement logic appears in three other inspected systems, but with different scientific meanings. NGC4088 is the clearest positive projection/history case: adding the independently motivated warp/history layer improves the replay, while the stronger additive-plus-clock stress curve is kept as a control because its source channels overlap under the source-token priority protocol of Subsection 2.6. NGC4013 is a guardrail case: moving from a simple compact proxy to a warp/vertical-overlay family improves the readout, but adding an extra Ξ_t overlay worsens the current proxy and therefore warns against double-counting the same projected vertical/warp evidence; by the same protocol, the vertical/warp source token is consumed by the mixed morphology kernel unless an independent clock/readout separator is supplied. NGC7331 is a caveated source-sharpening case: a broad outer-warp window already improves over the simple carrier, and a sharper source-window replay gives a small additional gain, but the broad-window uncertainty remains part of the accepted interpretation.

Galaxy	Source-frozen replay step	RMSE	Interpretation
NGC4088	Base projection	11.62	Source-supported projection control.
NGC4088	Additive warp/history	9.39	Accepted projection/history refinement.
NGC4088	Clock-only control	10.50	Clock replay retained as a control, not an endpoint.
NGC4088	Additive plus clock stress	8.38	Lower RMSE, but source-channel overlap keeps it as a stress control.
NGC4013	Compact proxy	16.99	Original simple morphology proxy.
NGC4013	Warp/vertical overlay	11.45	Source-supported mixed morphology refinement.
NGC4013	Exponential-disk plus WVO protocol	10.61	Prospective source-frozen protocol reference.
NGC4013	Ξ_t overlay control	12.04	Double-counting guardrail; worsens the current proxy.
NGC7331	Exponential-disk carrier	23.47	Simple broad projected-disk carrier.
NGC7331	V1 broad outer warp	22.26	Accepted broad-window reference.
NGC7331	V2 fractional onset	22.73	Control replay; not the best source sharpening.
NGC7331	V3 source-sharpened	22.13	Small source-sharpened outer-warp gain.
NGC7331	Wrong sharpened control	22.91	Wrong-projection control, worse than the source-sharpened replay.

Table 10: Additional morphology-refinement patterns. Entries are RMSE values in $\mathrm{km\,s^{-1}}$. These rows are not an independent validation sample. They show how a residual-blind source review can sharpen a readout family, and also how the protocol blocks over-layering when a projection/time channel would double-count the same source evidence. The generating artifact is `data/derived/additional_morphology_refinement_ladders_summary.csv`.

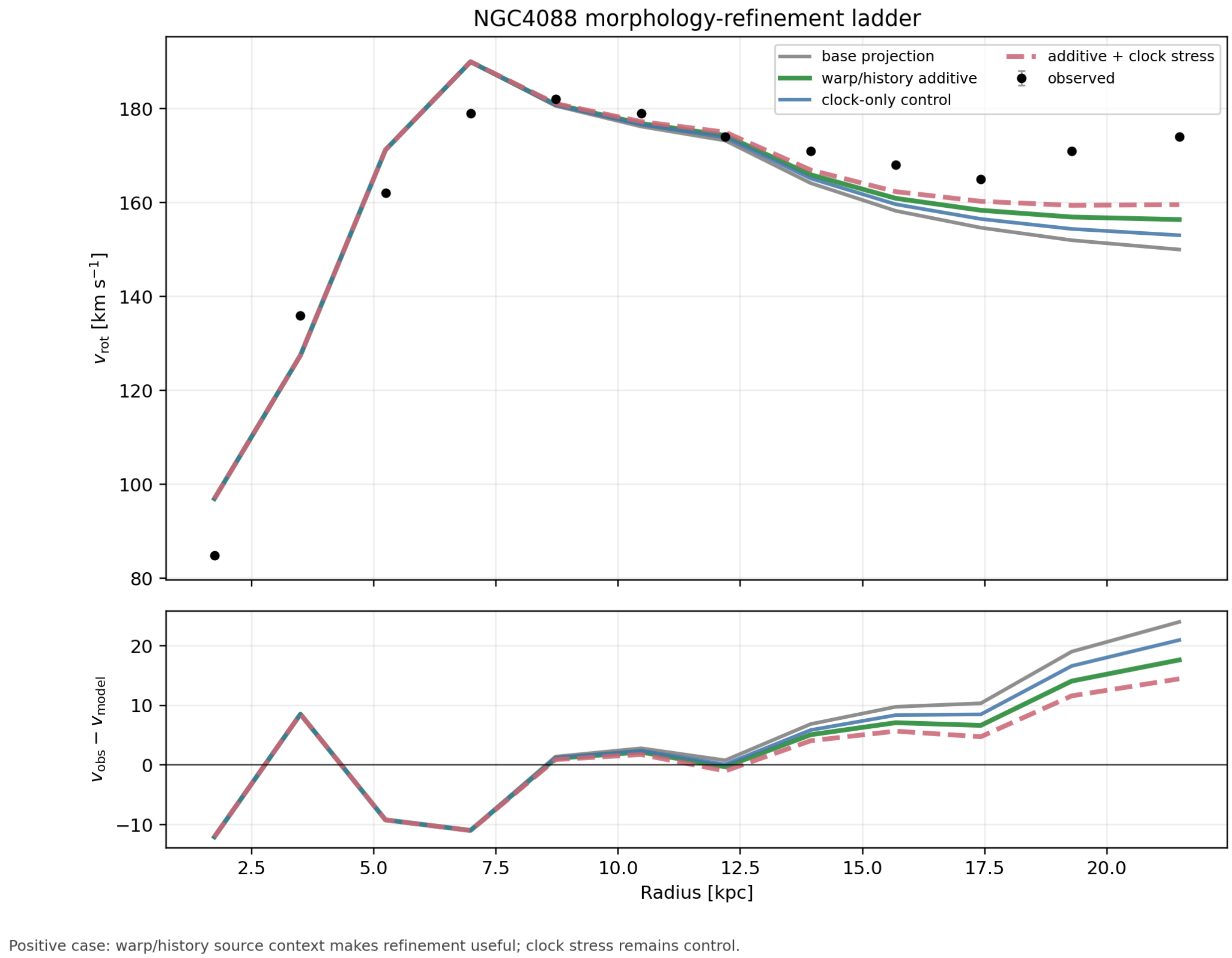


Figure 14: NGC4088 projection/history refinement ladder. The source-supported warp/history layer moves the Tau Core replay closer to the measured rotation curve. The additive-plus-clock stress curve is shown because it lowers RMSE further, but it is not promoted to endpoint validation: it overlaps source channels and therefore remains a control.

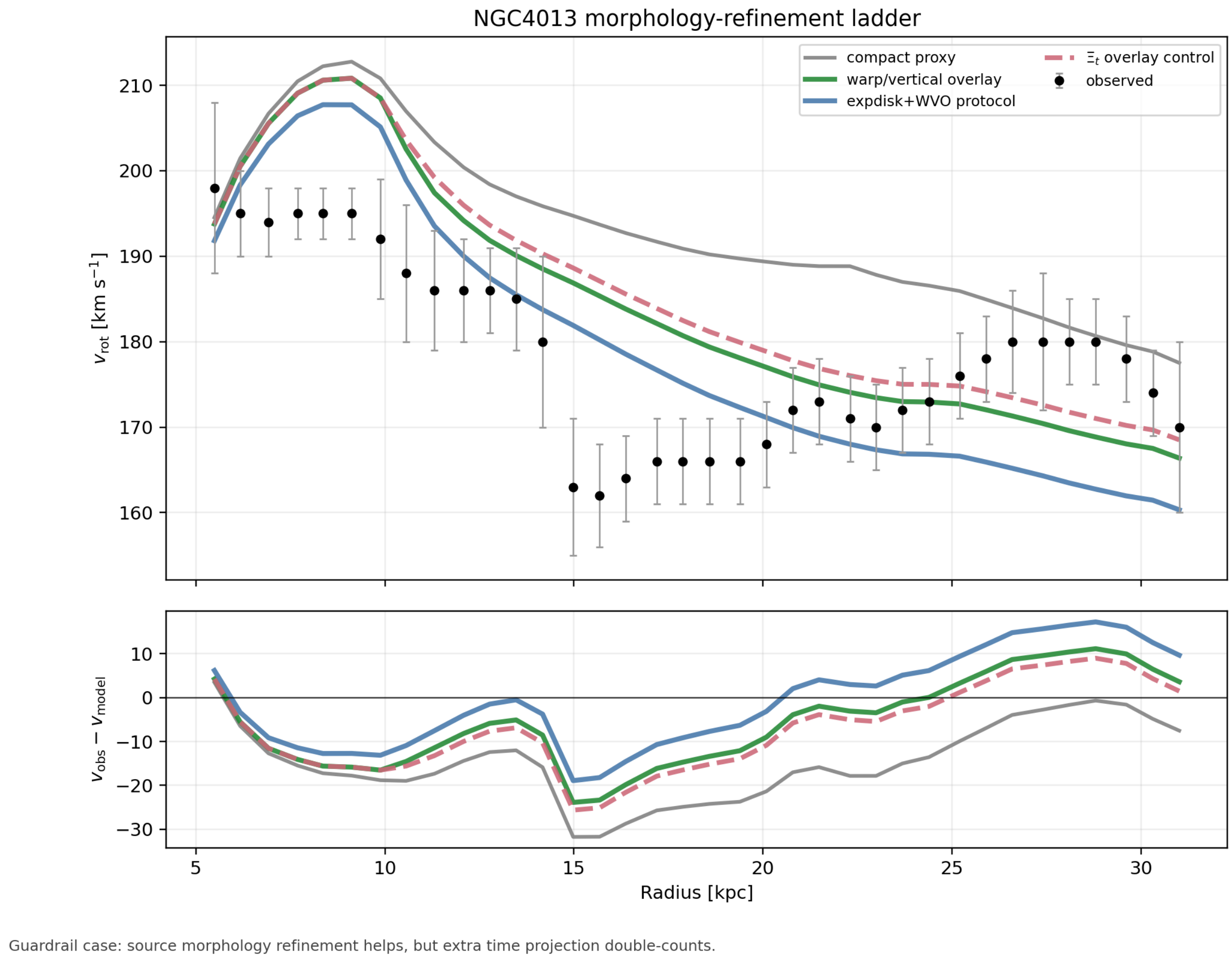


Figure 15: NGC4013 morphology-refinement guardrail. The source-supported warp/vertical-overlay route improves over the simple compact proxy, and the prospective exponential-disk plus WVO protocol is closer still. The Ξ_t overlay control worsens the replay, illustrating that the projection/time layer is not a universal improvement term and must not double-count the same vertical/warp evidence.

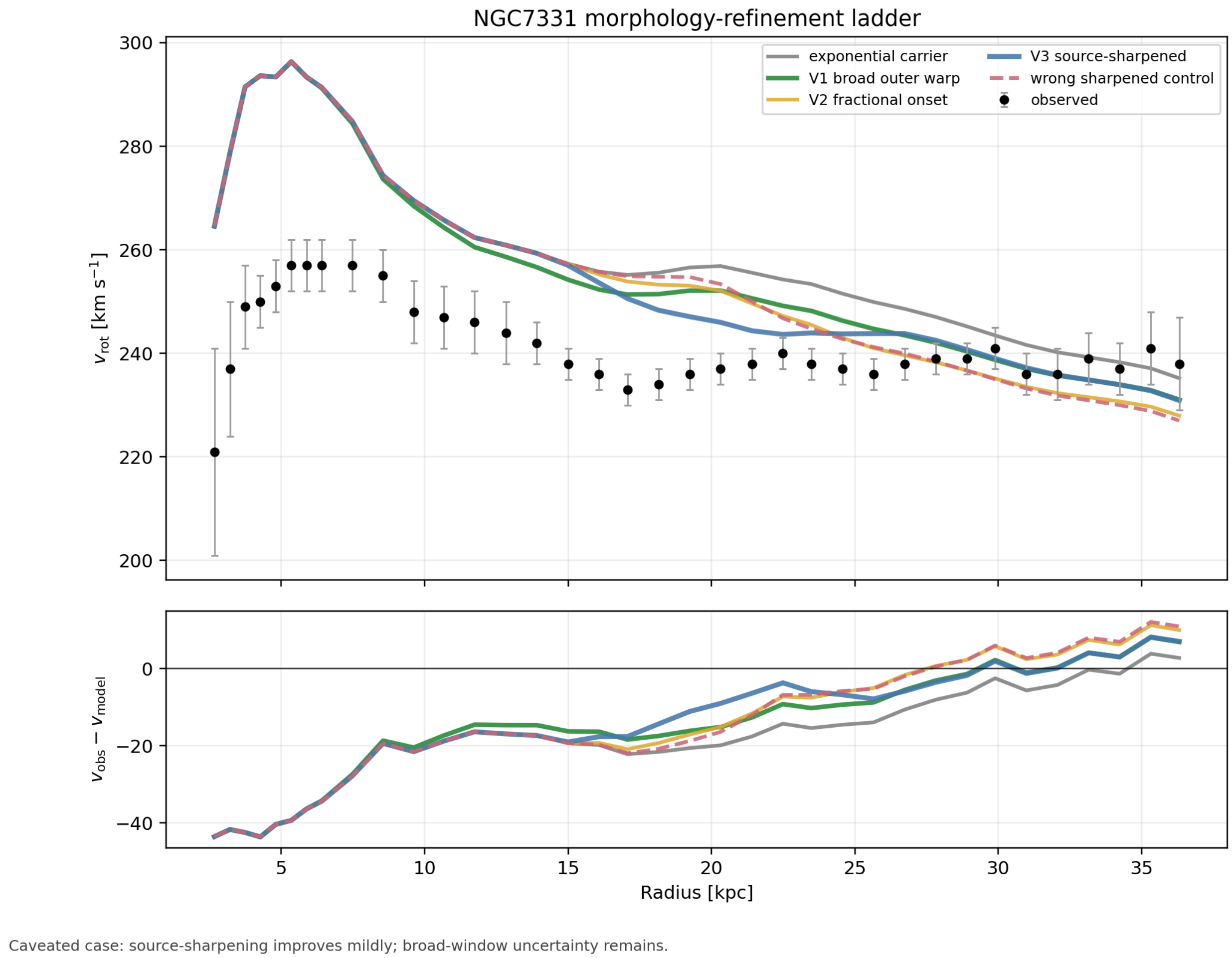


Figure 16: NGC7331 source-sharpening replay. The broad outer-warp window gives a modest gain over the exponential carrier, while the source-sharpened replay is slightly better and the wrong-projection sharpened control is worse. The case remains caveated because the accepted route still depends on a broad vertical/outer-warp window.

Together these refinement ladders clarify the paper's intended empirical logic. A more detailed morphology description is allowed to improve the readout only when it is frozen from source-side information before scoring. When the extra layer merely overlaps an already-used source channel, the protocol keeps it as a control even if its RMSE happens to improve. The scientific signal is therefore not "add more terms until the curve follows the data", but "source-complete morphology selects a more appropriate readout family, and inappropriate extra channels are preserved as controls".

8.3 Morphology adequacy review for NGC4088 and NGC4013

Because NGC4088 and NGC4013 carry much of the visible refinement narrative, we also record a separate source-side morphology adequacy review. This review does not read endpoint residuals. It asks only whether the assigned readout-relevant morphology is source-supported and whether it is precise enough for the role claimed in this paper. The full ledger is released as `ngc4088_ngc4013_morphology_adequacy_review_*` under `data/derived/`, with the markdown report under `reports/`.

NGC4088. The $K_{\mathrm{warp/history}}$ direction is source-supported. The distorted, asymmetric warped H I disk and companion/history context strongly support the warp/history classification, so the family is not being inferred from the rotation residual. However, the numeric morphology precision is still open: independent acceptance of x_{warp}, q_{warp}, memory/history, and the $\epsilon_{\mathrm{cross}}$ bound is required before this becomes a source-complete endpoint-grade morphology label.

NGC4013. The mixed exponential-disk plus warp/vertical-overlay direction is source supported at caveated prospective-protocol level. The compact lane is source-rejected, while smooth edge-disk, warp/flare, vertical component, and lag-context evidence support the mixed-overlay classification. The result is not retroactive validation, and the Ξ_t overlay remains a double-counting control under the source-token priority protocol unless new non-overlap evidence is supplied.

In both rows the adequacy review uses source ledgers only and explicitly forbids observed-velocity or residual inputs. Its function is to prevent the paper from treating curve improvement itself as morphology evidence.

A follow-up source-strengthening packet sharpens this verdict. For NGC4088, Verheijen and Sancisi's H I atlas directly supports a strongly distorted disk, a strong position–velocity asymmetry, an asymmetric warp, side-dependent position-angle change, and nearby companion/history context [19]. This upgrades the broad warp/history class to strong qualitative source support, but it still does not accept the numeric x_{warp}, q_{warp}, memory/history, or $\epsilon_{\mathrm{cross}}$ fields. For NGC4013, the source situation is more numeric: the H I modelling gives a line-of-sight warp onset near $1.2' \simeq 10$ kpc, a warp orientation of about 70° from the line of sight, model scale heights of 500, 350, and 210 pc across the one-component, warp, and warp-plus-flare models, and a lag that shallows from $-35\ \mathrm{km\,s^{-1}\ kpc^{-1}}$ at 5.8 kpc to zero near R_{25} [21]. The Spitzer vertical decomposition also supports an extra flattened component with $z_{\mathrm{EC}} \sim 3$ kpc and roughly 20–26 per cent of the disk mass [3]. Thus NGC4013 is strengthened at source-numeric mixed-overlay level, while NGC4088 is strengthened at broad class level only. The corresponding packet is `ngc4088_ngc4013_strengthened_source_review_*` under `data/derived/`.

We therefore reran a separate score-consequence audit with the source review held fixed. For NGC4013, the source-supported refinement gives the expected directional improvement: the original compact proxy has RMSE $16.99\ \mathrm{km\,s^{-1}}$, the warp/vertical-overlay route has $11.45\ \mathrm{km\,s^{-1}}$, and the prospective exponential-disk plus WVO frozen protocol has $10.61\ \mathrm{km\,s^{-1}}$. The last number is the strongest NGC4013 score in this packet, but it remains prospective/protocol-only rather than retroactive endpoint validation. For NGC4088, the accepted warp/history route already has RMSE $11.62\ \mathrm{km\,s^{-1}}$, beating the best local baseline ($25.40\ \mathrm{km\,s^{-1}}$) and all wrong-family controls. The additive, clock-only, and additive-plus-clock control replays can reach 9.39, 10.50, and $8.38\ \mathrm{km\,s^{-1}}$, respectively, but those improvements remain controls because the source-strengthening packet has not promoted new endpoint-grade x_{warp}, q_{warp}, memory/history, or $\epsilon_{\mathrm{cross}}$ inputs that survive the Subsection 2.6 T/A test. The score ledger is released as `ngc4088_ngc4013_strengthened_scoring_audit_*`.

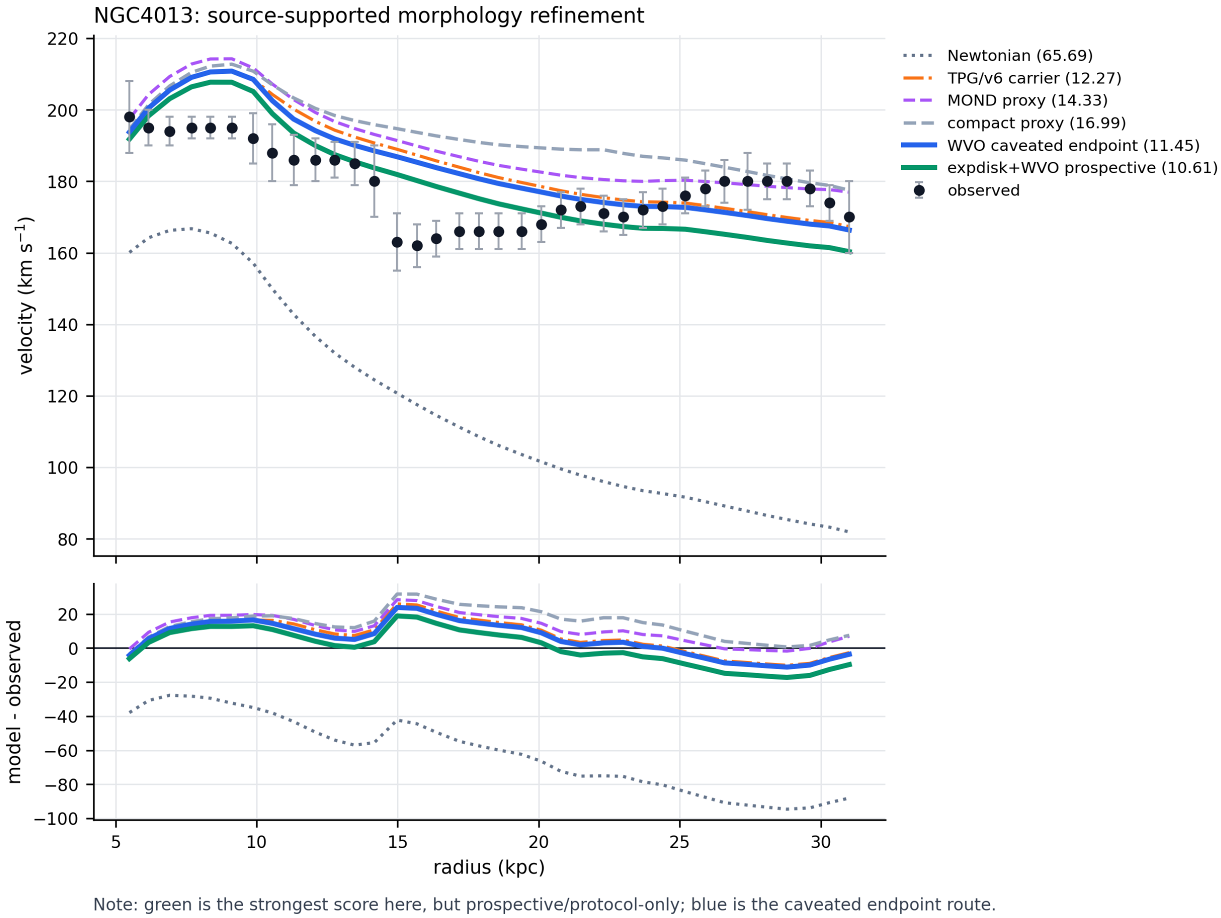


Figure 17: NGC4013 strengthened rotation comparison. The source-rejected compact proxy is shown against the source-supported warp/vertical-overlay route, the prospective exponential-disk plus WVO protocol, and the local Newtonian, TPG/v6, and MOND comparators. The green curve gives the lowest RMSE in this packet, but remains prospective/protocol-only; the blue curve is the caveated endpoint route.

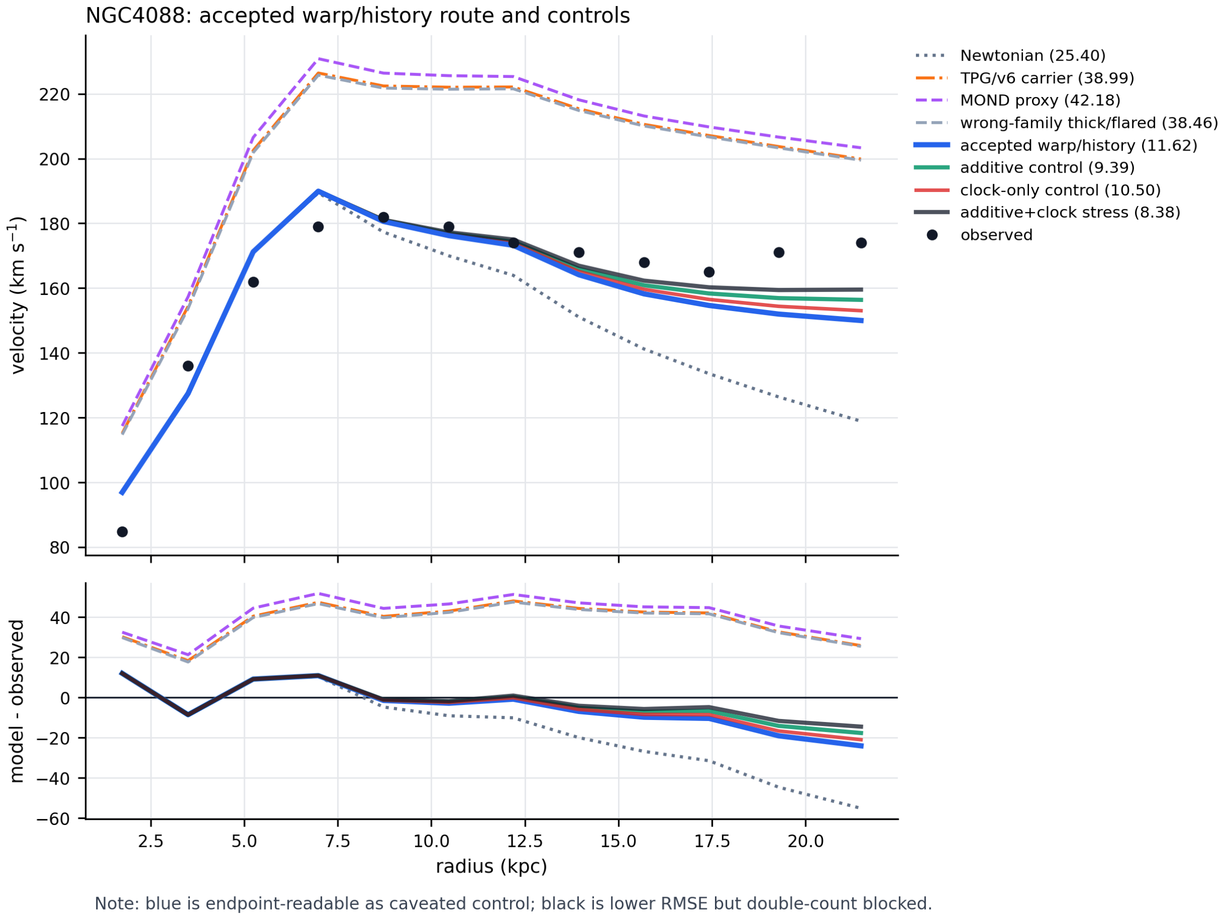


Figure 18: NGC4088 strengthened rotation comparison. The accepted warp/history route is compared with Newtonian, TPG/v6, MOND, a wrong-family thick/flared control, and additive/time projection controls. The black stress curve has the lowest RMSE, but remains double-count blocked; the blue warp/history route is the endpoint-readable caveated control.

The double-count statement for NGC4088 is not only a verbal caveat. We also release an explicit non-overlap certificate implementing the source-token priority protocol of Subsection 2.6. Let A denote the source subspace already assigned to the accepted additive warp/history route, generated by warp geometry, radial onset, q_{warp} source strength, and morphology-history phase. Let T denote the current Ξ_t source ledger. The ledger contains four terms: f_{PA}, f_R, f_q, and f_{mem}. The certificate maps these respectively to shared warp geometry, shared radial-onset support, direct q_{warp} source-strength overlap, and shared morphology-history phase. Hence all current Ξ_t generators lie in A, and the endpoint-admissible clock load is the quotient load in T/A. Numerically the raw clock load is 1.375, but the orthogonal load is zero. Therefore the accepted combined route must set $\Xi_{\mathrm{eff}} = 1$, and the lower-RMSE additive-plus-clock curve remains a stress/control curve rather than endpoint evidence. The certificate is released as `ngc4088_time_projection_nonoverlap_certificate_*`.

9 Per-kernel derivation ledger

Table 11 makes explicit how the main kernels used in this paper are obtained. The point of the table is not to add new results; it is to expose the derivation chain that is otherwise spread across source-review, formula-freeze, and scoring artifacts. A row is endpoint-readable only when the source label, radial support, amplitude or carrier rule, and non-overlap status are all frozen before scoring. Otherwise the row is retained as a control, diagnostic, candidate, or blocked route.

Table 11: Per-kernel derivation ledger for the concrete readout families used or proposed in Paper 2. "Formula role" gives the operational kernel shell, not a validated universal law. "Status" records the claim boundary used in the paper.

Kernel route	Source statement	Observable / frozen input	Formula role	Status / artifact family
Morphology-only family	Residual-blind projected morphology class such as scale-tail, compact, exponential, ring, or thick/flared disk	Accepted morphology label and class-specific radial support proxy	$\delta v_K^2 = A_K K_K(R)$	preflight baseline family; upgraded only when source manifest and amplitude freeze pass
Projection-enriched mixed family	Present morphology alone is insufficient because observer/path visibility, warp, flare, vertical overlay, or trajectory phase changes the effective readout	Channel components $K_j(R)$, visibility weights $w_j(R)$, and source-side amplitudes A_j	$\delta v_{\rm proj}^2 = \sum_j A_j w_j K_j$	general Paper 2 shell; requires source-token and non-overlap gates
NGC4013 warp/vertical overlay	Smooth edge-on disk plus warp/flare, vertical component, and lag context	Warp onset near $1.2'$, warp orientation, scale-height models, lag profile, and Spitzer extra component	mixed warp/vertical-overlay radial gate added to an exponential-disk carrier	caveated endpoint/protocol route; Ξ_t overlay is a double-counting control
NGC4088 warp/history	Distorted, asymmetric warped H I disk with companion/history context	Broad warp/history support, side-dependent position-angle change, asymmetric H I morphology, and memory/history proxy	additive warp/history morphology route with $\Xi_{\rm eff} = 1$ after non-overlap audit	caveated accepted control route; additive-plus-clock remains stress/control because $T/A = 0$
NGC7331 outer-warp sharpening	Projected disk with caveated vertical/outer-warp structure	Broad outer-warp window and source-sharpened radial onset controls	outer-warp / vertical support window applied to projected-disk carrier	caveated source-sharpening route; not a final source-complete endpoint
NGC5907 edge-on projection	Highly edge-on projected disk where observer/path projection is already strong	Edge-on geometry, warp/truncation context, and projection saturation flag	projection-saturated mixed kernel; richer path term only if new source data justify it	near-neutral/saturated control; no active extra channel without new evidence

Kernel route	Source statement	Observable / frozen input	Formula role	Status / artifact family
NGC0891 / NGC4217 EVH–DVH	Pure global beta transfer overshoots; source review supports edge-on vertical/dust/H I/radio-halo readout	Edge-on dust lanes, H I halo, extra-planar gas, radio/halo evidence, and disk scale R_s	$v^2 = v^2_{\rm carrier}[1+(\beta_{\rm cl}-1)K_{\rm EVH/DVH}(R)]$	source-blind morphology-refinement controls; ordered improvement, not population validation
UGC12506 $\Theta_{\rm morph}$	Late-settling / projection-history morphology phase is plausible but not sufficient as a clock endpoint	Outer radial shape and source-status load assigned to morphology-state phase	$v^2_\theta = v^2_{\rm projection/history} + A_\theta K_\theta(R)$	diagnostic/control; improves directionally but remains non-endpoint
UGC12506 Ξ_t interval	High-inclination, high-spin, extended H I envelope suggests a small clock/readout interval, with path term set to zero	Source load L, protocol cap $\epsilon_{\rm cap} = 0.035$, and frozen ϵ_t interval	$\Xi_t(R) = 1 + \epsilon_t K_t(R)$, with $\epsilon_t = \epsilon_{\rm cap} L/(1+L)$	caveated interval/control manifest; endpoint blocked until K_t and cap origin are accepted
UGC12506 mass/envelope closure	Source-native NFW/high-spin envelope route has good shape but missing normalization	Spin proxy, NFW/HSE preference, inclination, and selected carrier route	$v^2 = \beta_{\rm cl} v^2_{\rm carrier}$, with $\beta_{\rm cl}$ source-derived but post-diagnostic in this draft	strong post-diagnostic source candidate; requires independent transfer before endpoint use
IC4202 smooth-disk projection prekernel	Fresh projection-rich smooth edge-on candidate has a source-supported active branch but missing value freeze	Inclination and disk support window $W_{\rm disk}(R; R_s, R_{\rm HI})$	$K_{\rm pre}(R) = \sin^2 i\, W_{\rm disk}(R; R_s, R_{\rm HI})$	formula/kernel blocked; needs $q_{\rm reg}$, $z_{\rm cl}$, and $e_{\rm proj}$ freeze

This ledger is deliberately asymmetric: some rows are formula shells, some are control kernels, and some are blocked future routes. That asymmetry is a feature of the protocol. The paper does not claim that every listed kernel is validated. It claims that when a kernel is used, the path from source morphology to radial formula, amplitude/carrier rule, and endpoint status is auditable.

10 Problematic-case route ledger

10.1 Candidate projection channels for problematic cases

The same audit also tells us which additional projection channels are worth trying next, and which ones should remain inactive. Table 12 is a route ledger, not a score table. It records the most plausible Tau-morphology-state projection channel for each problematic or caveated galaxy, together with the immediate guardrail. No row in the table authorizes an endpoint score by itself.

Galaxy	Current role	Priority channel	Required next gate
UGC12506	primary stress case, still underpredicted	mass/envelope plus metric-closure readout; observer/path and clock remain secondary controls	derive a source-frozen mass/envelope–closure kernel and ablate it against the Θ_{morph} and Ξ_t-cap controls
NGC4088	improving warp/history/asymmetry case	trajectory/phase and asymmetry/history readout; clock only as control	keep the accepted additive warp/history route with $\Xi_{\mathrm{eff}} = 1$; reopen clock only with non-overlapping evidence that survives the T/A source-token priority test
NGC4013	worsened mixed-overlay case	mixed warp/vertical-overlay readout; no current Ξ_t promotion	preserve as a retrospective mixed-reference control unless a fresh analogue or independent clock observable is found
NGC7331	worsened broad outer-warp case	sharpened vertical/outer-warp readout; metric/closure only after window sharpening	freeze a narrower outer-warp/vertical window before adding any clock or closure layer
NGC5907	near-neutral projection-saturated case	observer/path edge-on projection and warp/truncation	treat as projection-saturated unless new path or vertical profile data justify a richer channel
NGC4183	weak/null projection control	quiet weak-projection limit	retain as null control; do not introduce active projection channels without new source evidence

Table 12: Candidate projection-channel ledger for problematic and caveated galaxies. The table records the next plausible source-side readout channel, not an endpoint promotion. Every listed new channel remains gated by source-freeze, non-overlap, and ablation tests before scoring.

We also convert this ledger into an executable next-gate CSV artifact. The artifact is explicitly marked as a not-endpoint gate plan; it contains six galaxies and authorizes zero endpoint promotions. Two rows have control/replay paths: UGC12506 proceeds only to a source-native NFW/HSE mass-envelope plus metric-closure ablation against the Θ_{morph}-only, $\Theta_{\mathrm{morph}} + \Xi_t$-cap, and source-envelope controls; NGC4088 keeps the accepted additive warp/history route while clock readout remains a control. The other rows stay inactive or blocked: the current NGC4013 Ξ_t proxy is rejected as double-counting, NGC7331 needs source-sharpened outer-warp/vertical windowing before refined replay, NGC5907 is projection-saturated at the current proxy level, and NGC4183 remains the weak/null projection control.

11 Predeclared projection-enriched population-validation gate

The current audit motivates a catalogue test but does not pass one. A population claim requires a residual-blind projection-enriched catalogue in which source-frozen observer/projection labels select the enriched family before endpoint scoring. The released protocol artifacts define the next endpoints:

DELTA_ENRICHED_SIMPLER enriched matched RMSE versus the simpler morphology-only proxy;

DELTA_ENRICHED_WRONG enriched matched RMSE versus wrong projection-enriched families;

SHUFFLED_PROJECTION_LABEL_NULL matched enriched score versus shuffled observer/projection labels;

BASELINE_COMPETITIVENESS enriched matched readout versus Newtonian, MOND/RAR-like, TPG/v6-like, and RMOND-facing comparators;

TRAJECTORY_PHASE_ABLATION present-only, trajectory-enriched, and trajectory/phase-enriched morphology kernels compared as a staged ablation;

PATH_ENVIRONMENT_ABLATION source-only, source-plus-observer, and path-environment kernels compared as a staged ablation.

The present status is the blocked gate

`PROJECTION_ENRICHED_POPULATION_VALIDATION_BLOCKED`
`CATALOGUE_AND_WRONG_LABEL_GATES`.

The gate requires at least 12 source-rich catalogue cases and at least 8 fresh cases not used in the current candidate audit. Endpoint scores are not run in this manuscript. Diagnostic scores are not allowed as label inputs.

12 Catalogue expansion status

The population gate is blocked, but not empty. The current expansion ledger contains 18 projection-rich rows: the five inspected candidate cases, one weak/null projection control, one frozen-protocol transfer seed, one formula-blocked named candidate, and 10 additional fresh projection-catalogue queue objects selected from residual-blind orientation/projection manifests. Twelve rows are potential fresh catalogue cases, satisfying the numerical size target in principle, but they are not formula-frozen endpoint cases.

Quantity	Current	Required	Interpretation
Catalogue rows listed	18	12	enough rows to define an expansion queue
Fresh candidate rows	12	8	enough possible fresh cases in principle
Formula/kernel blocked rows	12	0	population validation remains blocked
Source/orientation blocked rows	12	0	source-side catalogue work is the main bottleneck
Wrong-label caveated seed rows	2	0	NGC5907 and NGC7331 need replay controls

Table 13: Projection-enriched population expansion status. These counts come from the expansion ledger and are preparation counts, not endpoint scores.

NGC3198 is now a useful transfer seed rather than a population-validation row: a source-derived constant-H HALOGAS/EPG frozen protocol has been scored, but the same galaxy was already diagnostically inspected, so it cannot be counted as a fresh endpoint validation. IC4202 is the most concrete formula-blocked fresh candidate: it has an edge-on smooth-disk projection shell, but the projection-load values and closure normalization are not yet frozen. The broader queue contains high-inclination or projection-ready SPARC objects such as UGC07151, UGCA444, IC2574, and UGC06818, but these still need source-native orientation promotion, morphology-trajectory acceptance, and formula freeze manifests before scoring.

The IC4202 blocker has now been reduced to a value-freeze obligation rather than a generic morphology uncertainty. A source-supported projection prekernel $K_{\mathrm{pre}}(R) = \sin^2 i\, W_{\mathrm{disk}}(R; R_s, R_{\mathrm{HI}})$ is available from residual-blind support and inclination data. The nonzero active branch remains blocked only because $q_{\mathrm{reg}}(R)$, $z_{\mathrm{cl}}(R)$, and the independent projection/load scalar e_{proj} are not source-frozen. Acceptable resolutions are a source-native H I regularity/closure profile or a Tau-side value theorem fixed before endpoint scoring. Reusing inclination as the independent projection load, or using the observed rotation residual to set these values, is explicitly forbidden.

Thus the next validation step is concrete: resolve at least eight fresh source, orientation, and trajectory-phase blockers; build formula-freeze manifests; and then run the matched-versus-simpler, wrong-enriched, shuffled projection-label, trajectory-phase ablation, and baseline endpoints. Until those gates pass, the correct status remains catalogue expansion, not population validation.

13 Conclusion

Projection-enriched kernels are the natural next step after morphology-only forward readouts. The inspected galaxies show that adding observer/projection and morphology-trajectory information can improve the matched kernel over a simpler one-dimensional proxy, sometimes enough to cross a practical baseline competitiveness threshold. UGC12506 is deliberately retained as a cautionary case: projection-history enrichment improves the Tau curve substantially relative to simpler source shells, but still does not reach the strongest listed MOND/TPG diagnostic curves. The reason this case improves is also physically interpretable inside the source ledger: high inclination, high spin, an extended H I envelope, and projection-history stress all point toward a non-negligible observer/projection and clock-readout correction. This supports kernel development, not population validation. The full-time morphology phase replay adds an important negative-control feature: the extra trajectory/phase layer is not a universal gain term. It improves the strongest warp/history/asymmetry case (NGC4088), where the source-side morphology already indicates an unsettled warp/history/asymmetry state; it is nearly neutral in the weak or already saturated projection cases, and slightly worsens two already close mixed/projection kernels. This pattern is consistent with a morphology/history-dependent time/projection readout correction rather than an unconstrained curve-fitting knob.

The next scientific step is a predeclared projection-enriched catalogue test: source-frozen observer/projection labels must select enriched kernels before endpoint scoring, trajectory/phase proxies must be frozen from source-side evidence before use, wrong enriched families and shuffled projection labels must be run as controls, and baseline comparisons must be reported without universal superiority claims unless the population endpoint actually passes. The expansion ledger shows that a plausible candidate queue exists, but source and formula-freeze blockers remain. A still fuller path-aware kernel requires source-observer light-bundle and metric-environment data that are not yet modeled here.

The paper therefore establishes three limited results. First, projection-enriched and time-readout layers can be made testable as source-frozen controls rather than residual-fitted corrections. Second, the current controls are selective: they improve disturbed or projection-rich cases but remain neutral or worsen in already saturated systems, which argues against a universal gain-knob interpretation. Third, the beta-closure expression is a transfer hypothesis generated by the UGC12506 stress test, not same-galaxy validation. Its evidential test is independent, predeclared transfer to other high-spin edge-on systems.

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